FIELD TRACE

AI markup assist for Bluebeam Revu 21 — sample output set

This set shows what Field Trace would produce from a batch of piping photos. Each sheet pairs the original field photo, carrying **editable markups** (standard PDF annotations: polylines, callouts, dimensions, revision clouds and a stamp, on named layers), with an **as-built isometric** drafted from the same traced geometry. The last page rolls the accepted markups up into a quantity takeoff.

The markups are a first pass. They are meant to be reviewed, corrected and signed off in Bluebeam Revu 21.11 by the person who does this work today. Anything under 70% confidence is dashed and flagged for verification. A photo that is too uncertain is held instead of auto-drafted (see FT-06).

Materials (carbon steel, stainless, copper, galvanized, composite / plastic such as PEX, CPVC, GRE/GRP and lined pipe) are **suggested** from visible cues and printed text, with lower confidence on painted or insulated pipe. OD can be estimated from a photo with a scale reference. Wall thickness cannot: any schedule shown is a suggestion to verify. Valves are identified by operator and stem-motion family only, and instruments by type only.

Bluebeam OCR can make text in a photo searchable, such as stencils, tags and stamps. Field Trace works from the pipe geometry itself: it traces runs, places symbols and dimensions, and drafts the drawing. Field Trace is a Mimetic Technology prototype. It is not a Bluebeam product.

PHOTO CREDITS

FT-01	Shell and Tube Heat Exchanger Piping 1	Ivangiesen	CC0 1.0
FT-02	Ventilation piping installed in the back of house area of the future LIRR passenger concourse (CM014B, 03-19-2019)	MTA Capital Construction Mega Projects	CC BY 2.0
FT-03	Insulating piping in the back of house area of the future LIRR passenger concourse (CM014B, 03-27-2019)	MTA Capital Construction Mega Projects	CC BY 2.0
FT-04	Installing copper piping that will connect to a heating and cooling unit in the LIRR passenger concourse (CM014B, 1-24-2019)	MTA Capital Construction Mega Projects	CC BY 2.0
FT-05	Mechanical room.jpg	P199 at English Wikipedia	CC BY 2.5
FT-06	Installing hangers for piping along the east service corridor of the future LIRR concourse (CM014B, 08-30-2018)	MTA Capital Construction Mega Projects	CC BY 2.0
FT-07	Travail du cuivre, de l'acier galvanisé et de l'acier noir (cropped)	Le plombier du désert	CC BY-SA 4.0
FT-08	Potrubí	Janchromý	CC BY-SA 4.0
FT-09	PEX piping in basement ceiling	Tomwsulcer	CC0 1.0
FT-10	Cpvc sprinkler rohr kanadische holzkabache	Achim Hering	CC BY 3.0
FT-11	Differential pressure cell transmitter	Bitjungle	CC BY-SA 4.0

All photos are from Wikimedia Commons (FT-07 cropped and resized). They are not client job photos. The markups were prepared to show the output; they are not live model results. Project names, dates, tags and confidence values are sample data.

SHEET INDEX

PG	SHEET	TITLE	LOCATION	MARKUPS	ISSUES	STATUS
1	COVER	Cover, index, credits				
2	FT-01	Plate HX-1 — tower / condenser water	Mech. Rm B-104 · North wall	30	2	Draft — for review
3	FT-02	HWS offset at CMU wall — rough-in	Level B1 · Corridor C, column line 4	23	2	Draft — for review
4	FT-03	Overhead mains — east service corridor	Level B1 · East service corridor	20	2	Draft — for review
5	FT-04	Domestic water drops at wall penetrations	Level 1 · Back-of-house wall W-12	28	3	Draft — for review
6	FT-05	LP steam header & pump set	Central plant · Pump room	20	1	Reviewed
7	FT-06	Ceiling space above corridor — low light	Level B1 · Corridor D ceiling	14	2	Needs attention
8	FT-07	Mixed materials — black steel, copper, galvanized	Prefab shop · mock-up wall	42	3	Draft — for review
9	FT-08	Process rack — stainless and jacketed lines	Process area · utility rack	24	3	Draft — for review
10	FT-09	PEX and copper — basement ceiling	Residential · basement ceiling	16	2	Draft — for review
11	FT-10	CPVC sprinkler pipe at a firestop sleeve	Wood-frame floor · penetration	5	1	Draft — for review
12	FT-11	Pressure transmitter on stainless impulse tubing	Instrument stand	7	2	Draft — for review
13	TAKEOFF	Quantity takeoff — all photos				

MARKUP LAYERS

Pipe Runs	Polyline along each pipe centerline, colored by system
Fittings	Elbows, tees, couplings, flanges, caps
Valves & Instruments	Valves (operator + family, never body style), instruments (type; model / range not assumed), equipment
Hangers & Supports	Hangers, strut, penetrations / firestop
Dimensions	Dimensions with basis noted (CMU coursing, pipe OD)
Issues	Revision clouds with comments (punch / QA)
Labels	System + size + suggested material tags, pipe markers read, stamp

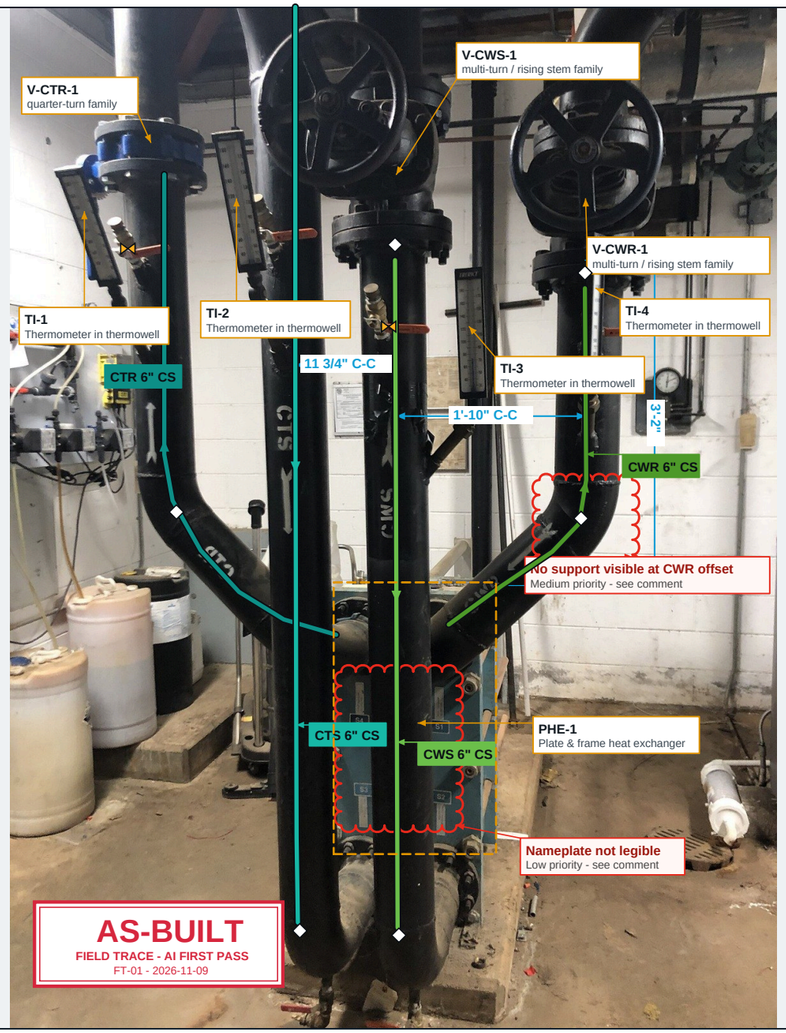
SYSTEM LEGEND

CTS	Cooling Tower Supply
CTR	Cooling Tower Return
CWS	Condenser Water Supply
CWR	Condenser Water Return
CHWS	Chilled Water Supply
CHWR	Chilled Water Return
HWS	Heating Hot Water Supply
HWR	Heating Hot Water Return
DCW	Domestic Cold Water
DHW	Domestic Hot Water
LPS	Low Pressure Steam
FP	Fire Protection
UNK	Unidentified — verify

FIELD TRACE

ANNOTATED FIELD PHOTO + AS-BUILT ISOMETRIC · FT-01

MIMETIC TECHNOLOGY PROTOTYPE · **SAMPLE DATA — NOT FOR CONSTRUCTION**



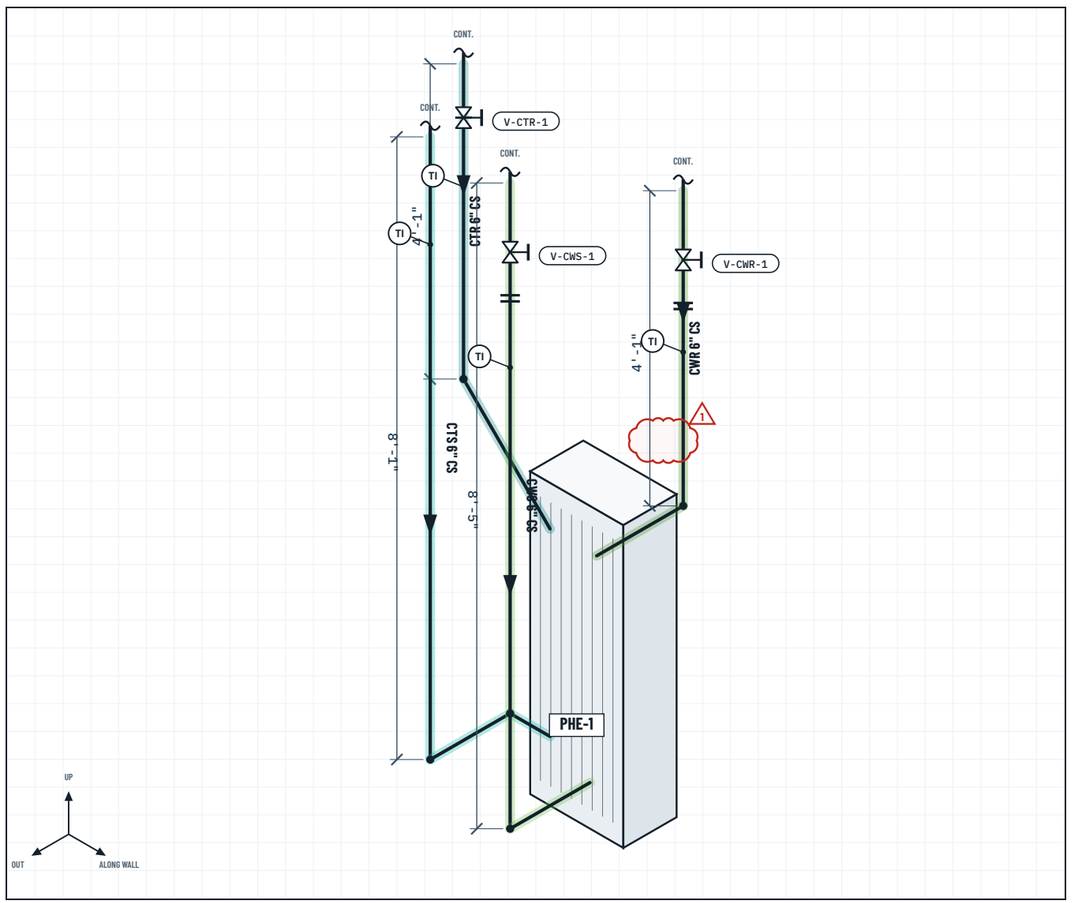
FIELD PHOTO FT-01 · MECH. RM B-104 · NORTH WALL · 2026-11-09 07:42

MARKUPS = EDITABLE PDF ANNOTATIONS · AUTHOR: FIELD TRACE AI

MARKUP SUMMARY — 30 MARKUPS · 2 ISSUES · 0 BELOW 70%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Polyline	CTR 6" CS	Pipe Runs	None	95%
Pipe Tag	CTR 6" CS	Labels	None	80%
Polyline	CTS 6" CS	Pipe Runs	None	93%
Pipe Tag	CTS 6" CS	Labels	None	80%
Polyline	CWS 6" CS	Pipe Runs	None	96%
Pipe Tag	CWS 6" CS	Labels	None	80%
Polyline	CWR 6" CS	Pipe Runs	None	94%
Pipe Tag	CWR 6" CS	Labels	None	80%
Handwheel valve	V-CWS-1	Valves & Instruments	None	92%
Handwheel valve	V-CWR-1	Valves & Instruments	None	91%
Gear-operated valve	V-CTR-1	Valves & Instruments	None	88%
Thermometer in thermowell	TI-1	Valves & Instruments	None	86%
Thermometer in thermowell	TI-2	Valves & Instruments	None	87%
Thermometer in thermowell	TI-3	Valves & Instruments	None	89%
Thermometer in thermowell	TI-4	Valves & Instruments	None	84%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Lever valve (drain / vent)	3/4" Lever valve (drain / vent)	Valves & Instruments	None	79%
Lever valve (drain / vent)	3/4" Lever valve (drain / vent)	Valves & Instruments	None	77%
Flange pair	6" Flange pair	Fittings	None	93%
Flange pair	6" Flange pair	Fittings	None	93%
45° elbow	6" 45° elbow	Fittings	None	90%
45° elbow	6" 45° elbow	Fittings	None	88%
90° LR elbow	6" 90° LR elbow	Fittings	None	71%
90° LR elbow	6" 90° LR elbow	Fittings	None	70%
Plate & frame heat exchanger	PHE-1	Valves & Instruments	None	90%
Dimension	1'-10" C-C	Dimensions	None	82%
Dimension	11 3/4" C-C	Dimensions	None	80%
Dimension	3'-2"	Dimensions	None	74%
Stamp	AS-BUILT — Draft	Labels	None	100%
Cloud+	No support visible at CWR offset	Issues	None	72%
Cloud+	Nameplate not legible	Issues	None	90%



AS-BUILT ISOMETRIC — PLATE HX-1 — TOWER / CONDENSER WATER

N.T.S. · DERIVED FROM PHOTO FT-01 · DIMENSIONS ESTIMATED

GENERAL NOTES

1. ALL PIPING 6" SCH 40 CS, 150# RF FLANGED U.N.O.
2. ELEVATIONS ESTIMATED FROM PHOTO — VERIFY IN FIELD
3. STENCILS READ: CTS · CTR · CWS · CWR

LEGEND

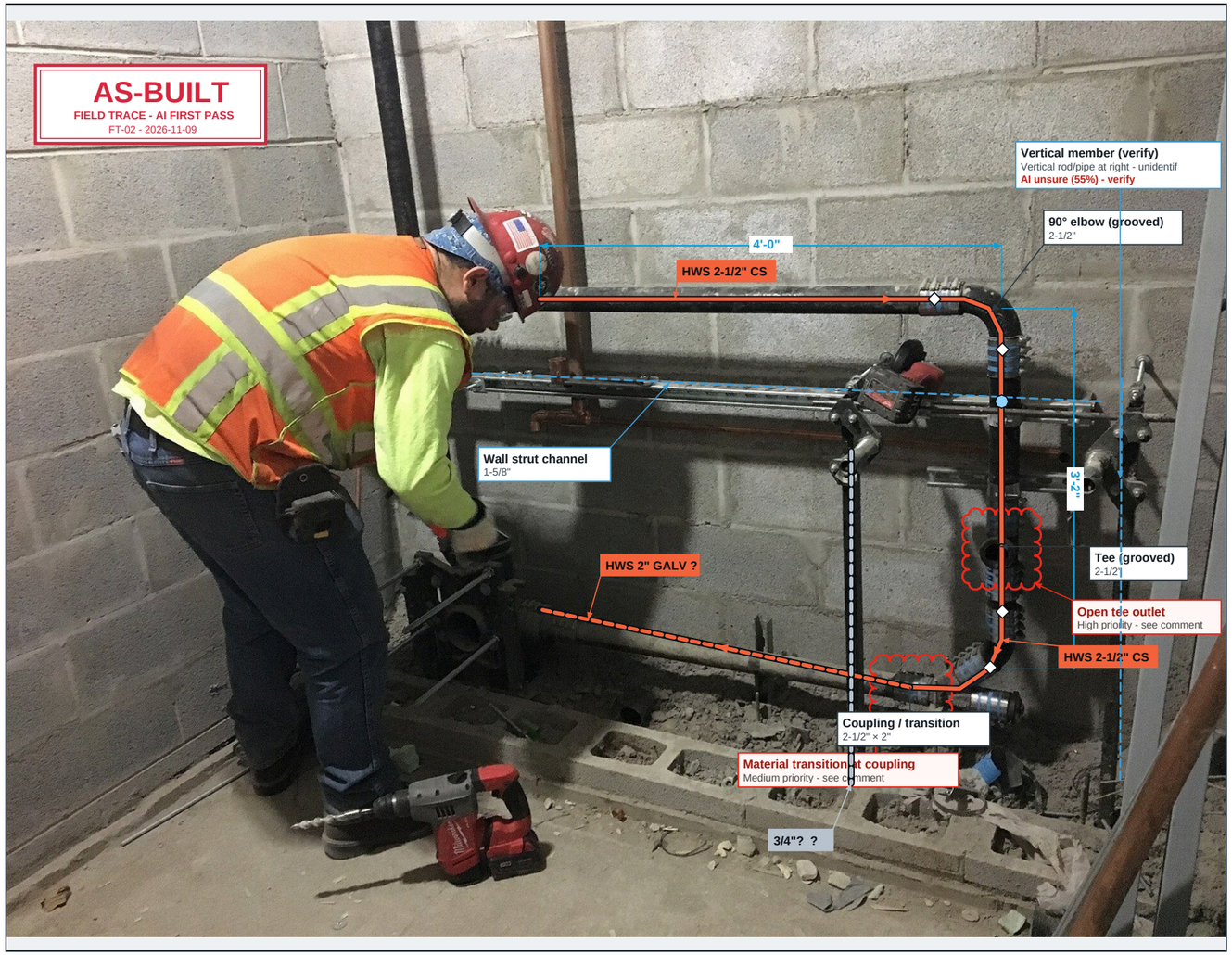
- CTR — Cooling Tower Return
- CTS — Cooling Tower Supply
- CWS — Condenser Water Supply
- CWR — Condenser Water Return
- CS — Carbon steel (sugg.)
- Below 70% — verify

REVISION CLOUDS

△1 No support visible at CWR offset (Medium)
△2 Nameplate not legible (Low)

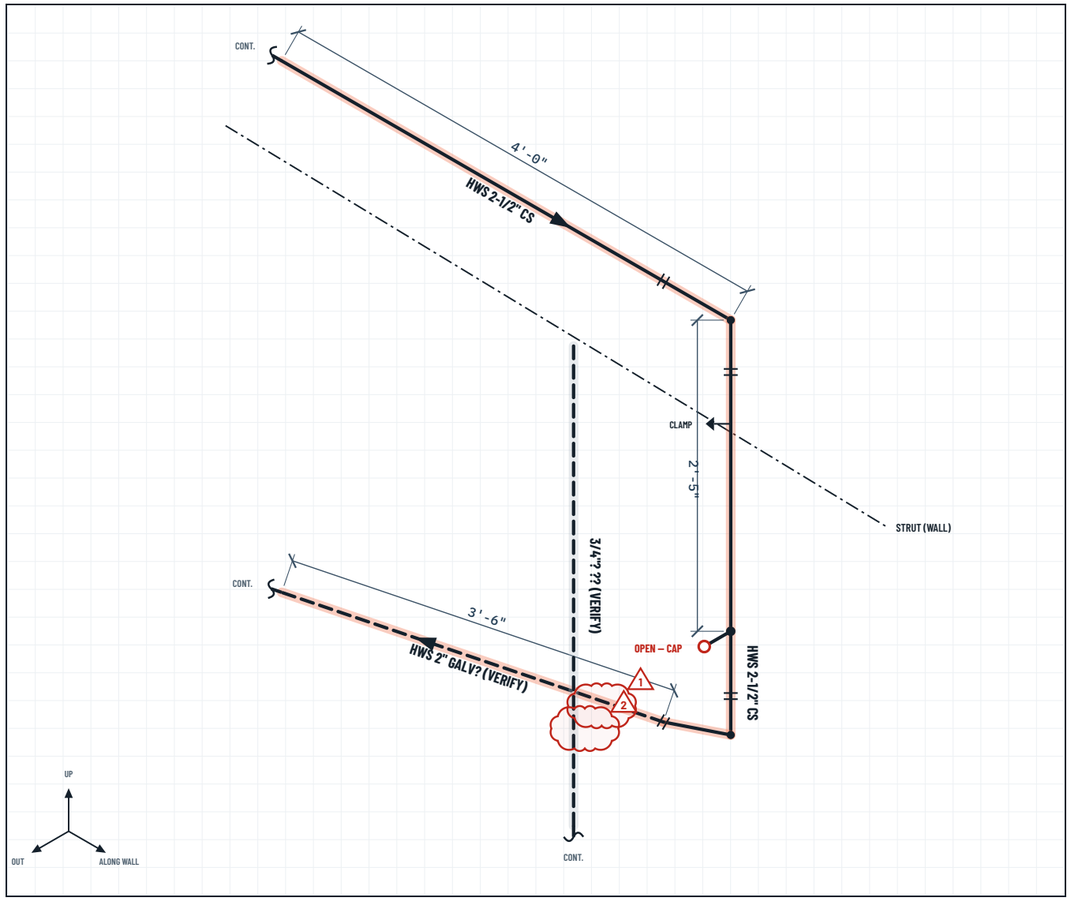
FIELD TRACE	MIMETIC TECHNOLOGY · MIMETICTECH.AI 80/20 SOFTWARE · PROTOTYPE			
PROJECT Sample Project — Mechanical Rough-In (DEMO-0142)			PREPARED BY MIMETIC TECHNOLOGY	
DRAWING TITLE AS-BUILT — PLATE HX-1 — TOWER / CONDENSER WATER			LOCATION Mech. Rm B-104 · North wall	
SOURCE PHOTO FT-01 · 2026-11-09 07:42	DRAWN FIELD TRACE AI	CHECKED —	SCALE N.T.S.	REV A
STATUS DRAFT — FOR REVIEW IN BLUEBEAM REVU 21.11			SHEET FT-01	

SAMPLE DATA · PUBLIC LICENSED PHOTO · NOT A CLIENT JOB PHOTO · NOT FOR CONSTRUCTION



FIELD PHOTO FT-02 · LEVEL B1 · CORRIDOR C, COLUMN LINE 4 · 2026-11-09 09:15

MARKUPS = EDITABLE PDF ANNOTATIONS · AUTHOR: FIELD TRACE AI



AS-BUILT ISOMETRIC — HWS OFFSET AT CMU WALL — ROUGH-IN

N.T.S. · DERIVED FROM PHOTO FT-02 · DIMENSIONS ESTIMATED

MARKUP SUMMARY — 23 MARKUPS · 2 ISSUES · 6 BELOW 70%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Polyline	HWS 2-1/2" CS	Pipe Runs	None	91%
Pipe Tag	HWS 2-1/2" CS	Labels	None	80%
Polyline	HWS 2-1/2" CS	Pipe Runs	None	86%
Pipe Tag	HWS 2-1/2" CS	Labels	None	80%
Polyline	HWS 2" GALV?	Pipe Runs	None	68%
Pipe Tag	HWS 2" GALV?	Labels	None	60%
Polyline	3/4"? ??	Pipe Runs	None	58%
Pipe Tag	3/4"? ??	Labels	None	50%
Rigid coupling	2-1/2" Rigid coupling	Fittings	None	93%
90° elbow (grooved)	2-1/2" 90° elbow (grooved)	Fittings	None	94%
Rigid coupling	2-1/2" Rigid coupling	Fittings	None	90%
Tee (grooved)	2-1/2" Tee (grooved)	Fittings	None	87%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Rigid coupling	2-1/2" Rigid coupling	Fittings	None	88%
90° elbow (grooved)	2-1/2" 90° elbow (grooved)	Fittings	None	83%
Coupling / transition	2-1/2" × 2" Coupling / transition	Fittings	None	70%
Wall strut channel	1-5/8" Wall strut channel	Hangers & Supports	None	92%
Pipe clamp at strut	2-1/2" Pipe clamp at strut	Hangers & Supports	None	80%
Vertical member (verify)	Vertical member (verify)	Hangers & Supports	None	55%
Dimension	4'-0"	Dimensions	None	81%
Dimension	3'-2"	Dimensions	None	78%
Stamp	AS-BUILT — Draft	Labels	None	100%
Cloud+	Open tee outlet	Issues	None	85%
Cloud+	Material transition at coupling	Issues	None	68%

GENERAL NOTES

1. 2-1/2" GROOVED STEEL, RIGID COUPLINGS U.N.O.
2. DIMENSIONS FROM CMU COURSING — ±2"
3. DASHED = BELOW CONFIDENCE THRESHOLD, VERIFY

LEGEND

- HWS — Heating Hot Water Supply
- CS — Carbon steel (sugg.)
- GALV — Galvanized steel (sugg.)
- ? — Not determinable (sugg.)
- Below 70% — verify

REVISION CLOUDS

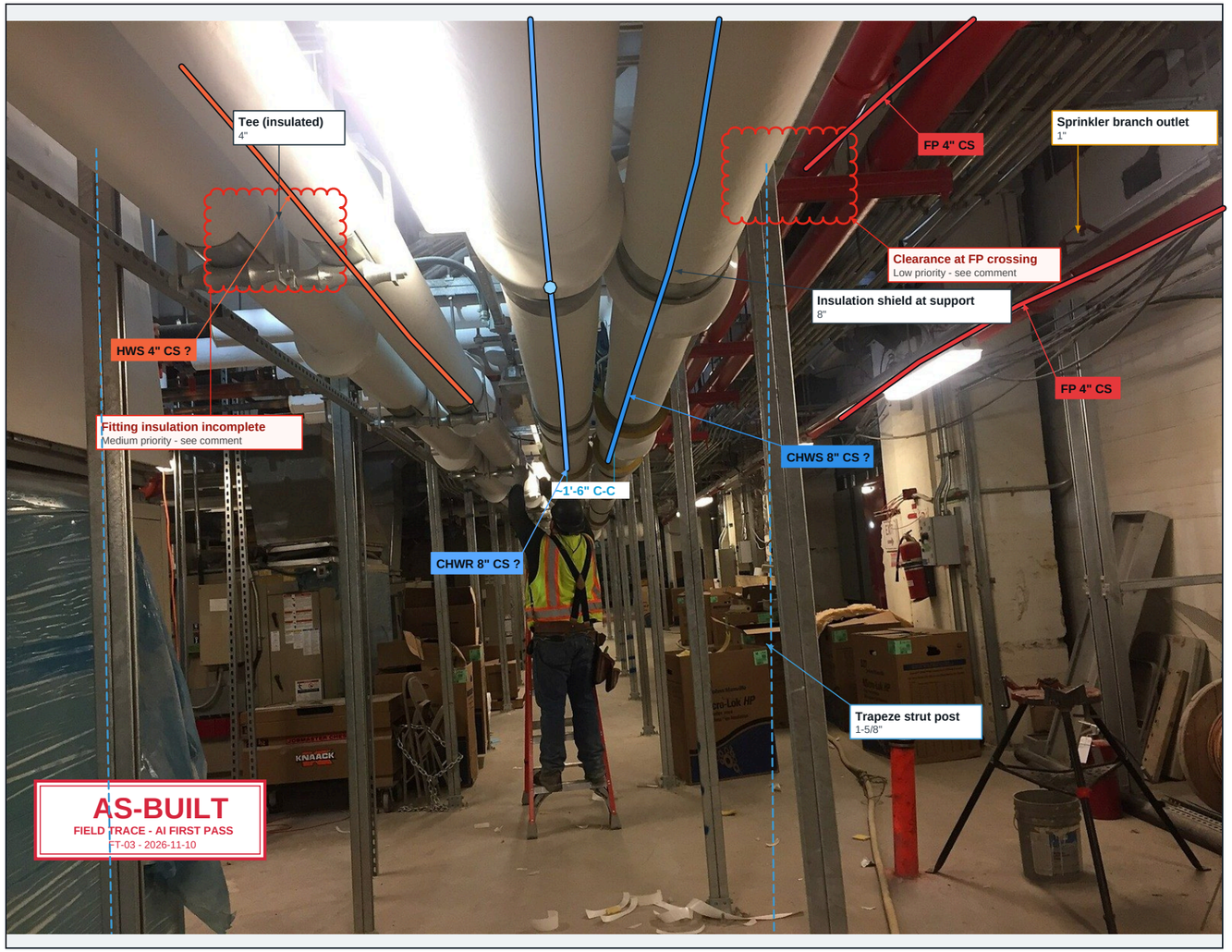
△1 Open tee outlet (High)
△2 Material transition at coupling (Medium)

FIELD TRACE

MIMETIC TECHNOLOGY · MIMETICTECH.AI
80/20 SOFTWARE · PROTOTYPE

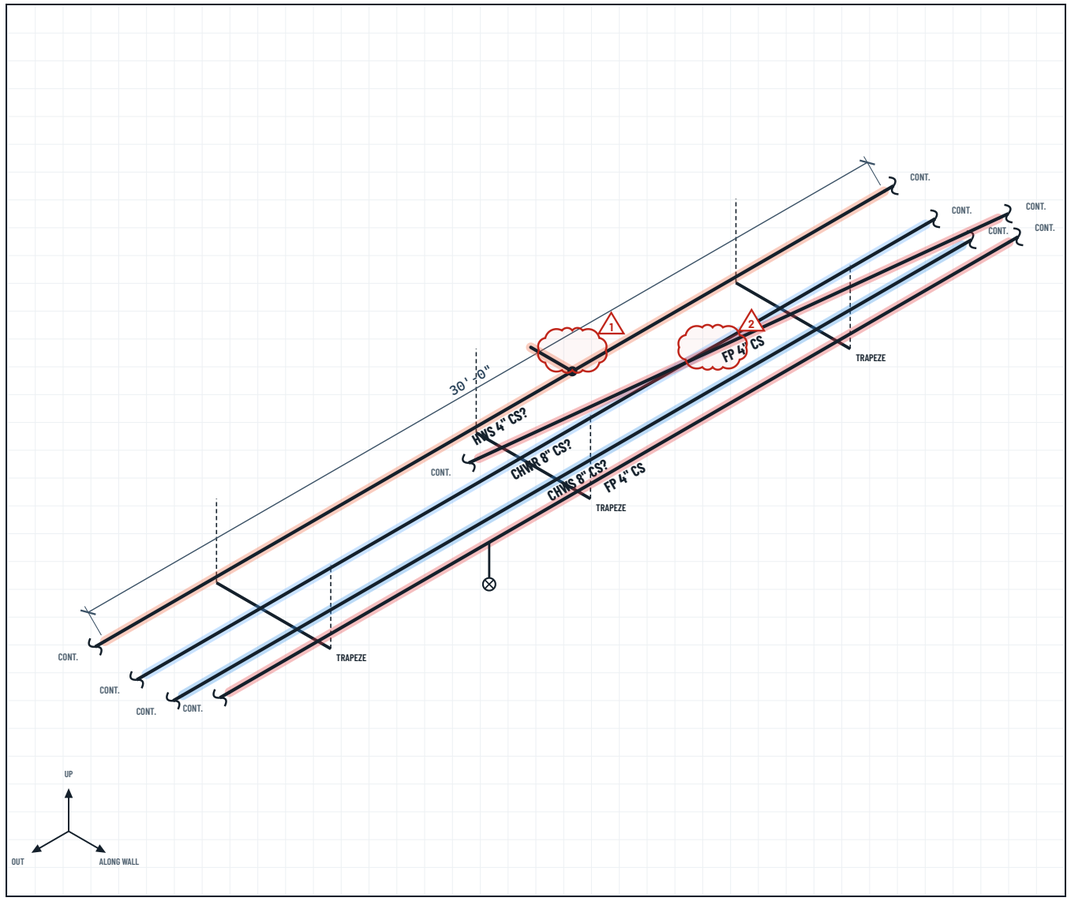
PROJECT: Sample Project — Mechanical Rough-In (DEMO-0142)			PREPARED BY: MIMETIC TECHNOLOGY	
DRAWING TITLE: AS-BUILT — HWS OFFSET AT CMU WALL — ROUGH-IN			LOCATION: Level B1 · Corridor C, column line 4	
SOURCE PHOTO: FT-02 · 2026-11-09 09:15	DRAWN: FIELD TRACE AI	CHECKED: —	SCALE: N.T.S.	REV: A
STATUS: DRAFT — FOR REVIEW IN BLUEBEAM REVU 21.11			SHEET: FT-02	

SAMPLE DATA · PUBLIC LICENSED PHOTO · NOT A CLIENT JOB PHOTO · NOT FOR CONSTRUCTION



FIELD PHOTO FT-03 · LEVEL B1 · EAST SERVICE CORRIDOR · 2026-11-10 13:05

MARKUPS = EDITABLE PDF ANNOTATIONS · AUTHOR: FIELD TRACE AI



AS-BUILT ISOMETRIC — OVERHEAD MAINS — EAST SERVICE CORRIDOR

N.T.S. · DERIVED FROM PHOTO FT-03 · DIMENSIONS ESTIMATED

MARKUP SUMMARY — 20 MARKUPS · 2 ISSUES · 5 BELOW 70%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Polyline	CHWS 8" CS?	Pipe Runs	None	88%
Pipe Tag	CHWS 8" CS?	Labels	None	50%
Polyline	CHWR 8" CS?	Pipe Runs	None	86%
Pipe Tag	CHWR 8" CS?	Labels	None	50%
Polyline	HWS 4" CS?	Pipe Runs	None	79%
Pipe Tag	HWS 4" CS?	Labels	None	50%
Polyline	FP 4" CS	Pipe Runs	None	90%
Pipe Tag	FP 4" CS	Labels	None	86%
Polyline	FP 4" CS	Pipe Runs	None	84%
Pipe Tag	FP 4" CS	Labels	None	84%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Insulation shield at support	8" Insulation shield at support	Hangers & Supports	None	83%
Insulation shield at support	8" Insulation shield at support	Hangers & Supports	None	80%
Trapeze strut post	1-5/8" Trapeze strut post	Hangers & Supports	None	86%
Trapeze strut post	1-5/8" Trapeze strut post	Hangers & Supports	None	84%
Sprinkler branch outlet	1" Sprinkler branch outlet	Valves & Instruments	None	74%
Tee (insulated)	4" Tee (insulated)	Fittings	None	70%
Dimension	~1'-6" C-C	Dimensions	None	62%
Stamp	AS-BUILT — Draft	Labels	None	100%
Cloud+	Fitting insulation incomplete	Issues	None	71%
Cloud+	Clearance at FP crossing	Issues	None	66%

GENERAL NOTES

1. RUN LENGTHS FROM ASSUMED 10'-0" SUPPORT SPACING
2. INSULATION: CHW 2", HW 1-1/2" (PER SPEC — VERIFY)
3. FP SHOWN FOR COORDINATION ONLY

LEGEND

- CHWS — Chilled Water Supply
- CHWR — Chilled Water Return
- HWS — Heating Hot Water Supply
- FP — Fire Protection
- CS — Carbon steel (sugg.)
- Below 70% — verify

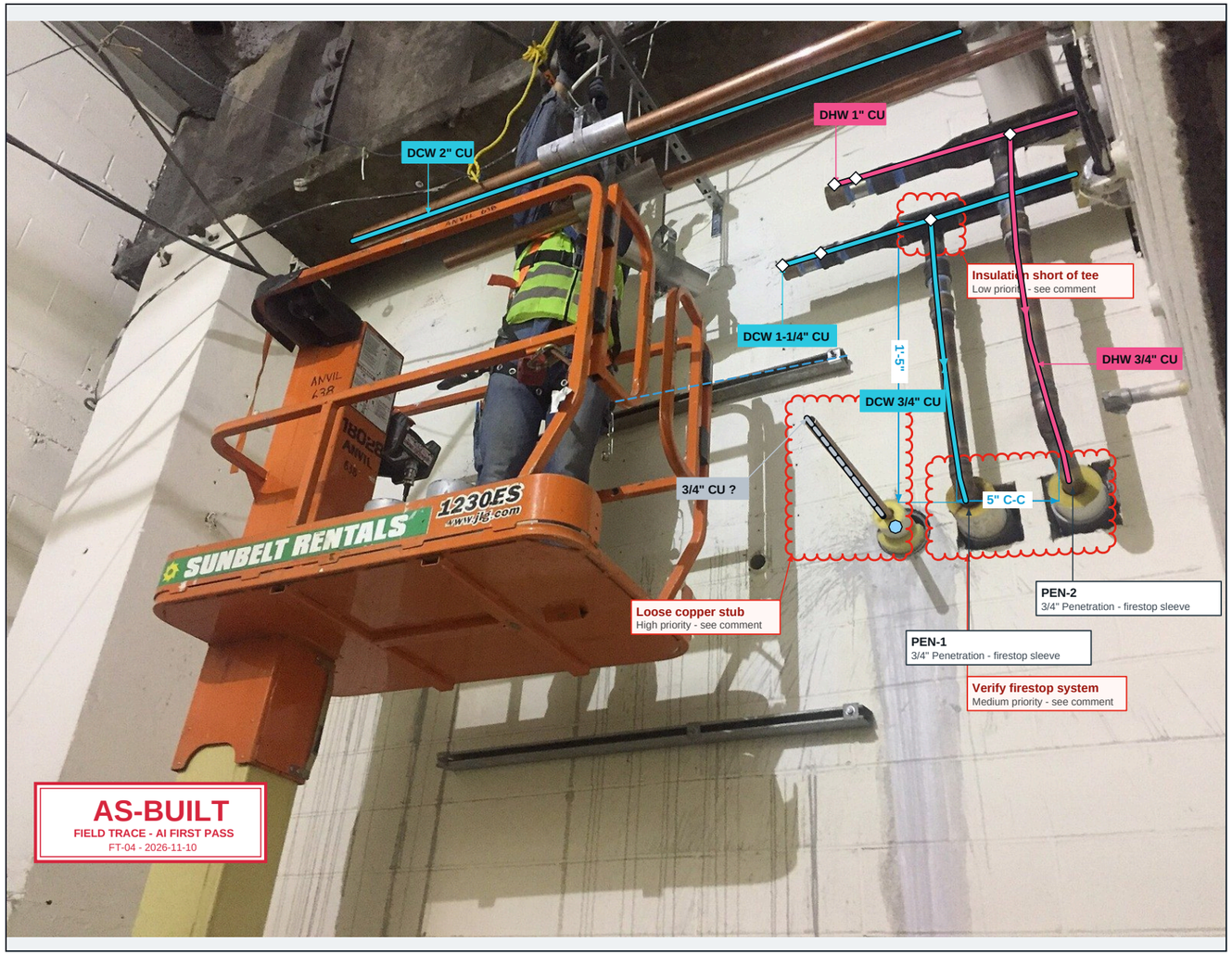
REVISION CLOUDS

△1 Fitting insulation incomplete (Medium)
△2 Clearance at FP crossing (Low)

FIELD TRACE

MIMETIC TECHNOLOGY · MIMETICTECH.AI
80/20 SOFTWARE · PROTOTYPE

PROJECT: Sample Project — Mechanical Rough-In (DEMO-0142)			PREPARED BY: MIMETIC TECHNOLOGY	
DRAWING TITLE: AS-BUILT — OVERHEAD MAINS — EAST SERVICE CORRIDOR			LOCATION: Level B1 · East service corridor	
SOURCE PHOTO: FT-03 · 2026-11-10 13:05	DRAWN: FIELD TRACE AI	CHECKED: —	SCALE: N.T.S.	REV: A
STATUS: DRAFT — FOR REVIEW IN BLUEBEAM REVU 21.11			SHEET: FT-03	



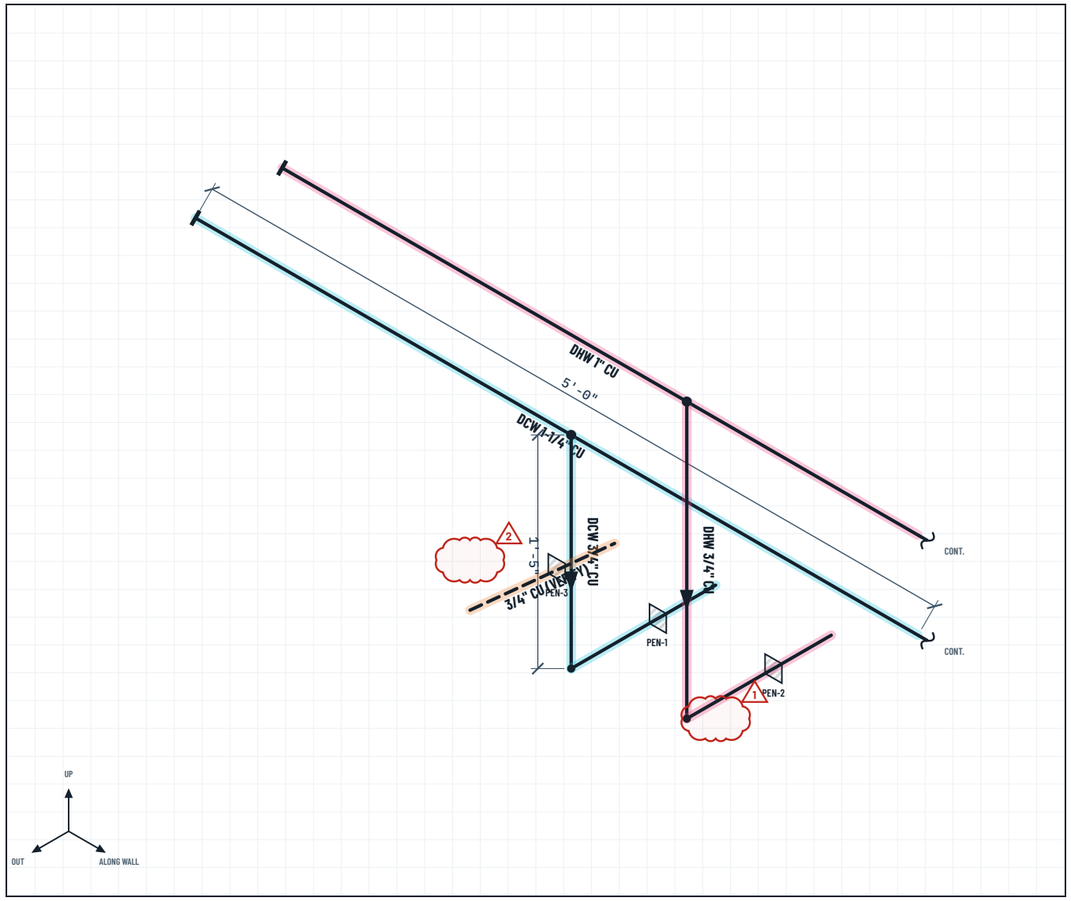
FIELD PHOTO FT-04 · LEVEL 1 · BACK-OF-HOUSE WALL W-12 · 2026-11-10 15:31

MARKUPS = EDITABLE PDF ANNOTATIONS · AUTHOR: FIELD TRACE AI

MARKUP SUMMARY — 28 MARKUPS · 3 ISSUES · 4 BELOW 70%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Polyline	DCW 1-1/4" CU	Pipe Runs	None	85%
Pipe Tag	DCW 1-1/4" CU	Labels	None	78%
Polyline	DHW 1" CU	Pipe Runs	None	83%
Pipe Tag	DHW 1" CU	Labels	None	78%
Polyline	DCW 3/4" CU	Pipe Runs	None	88%
Pipe Tag	DCW 3/4" CU	Labels	None	88%
Polyline	DHW 3/4" CU	Pipe Runs	None	76%
Pipe Tag	DHW 3/4" CU	Labels	None	76%
Polyline	DCW 2" CU	Pipe Runs	None	72%
Pipe Tag	DCW 2" CU	Labels	None	72%
Polyline	3/4" CU	Pipe Runs	None	60%
Pipe Tag	3/4" CU	Labels	None	60%
Tee	1-1/4" × 3/4" Tee	Fittings	None	84%
Tee (assumed)	1" × 3/4" Tee (assumed)	Fittings	None	66%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Press coupling	1" Press coupling	Fittings	None	82%
Press coupling	1-1/4" Press coupling	Fittings	None	82%
Cap (stub-out)	1" Cap (stub-out)	Fittings	None	75%
Cap (stub-out)	1-1/4" Cap (stub-out)	Fittings	None	75%
Penetration — firestop sleeve	PEN-1	Hangers & Supports	None	90%
Penetration — firestop sleeve	PEN-2	Hangers & Supports	None	90%
Penetration — firestop sleeve	PEN-3	Hangers & Supports	None	86%
Wall strut channel	1-5/8" Wall strut channel	Hangers & Supports	None	85%
Dimension	5" C-C	Dimensions	None	72%
Dimension	1'-5"	Dimensions	None	70%
Stamp	AS-BUILT — Draft	Labels	None	100%
Cloud+	Verify firestop system	Issues	None	80%
Cloud+	Loose copper stub	Issues	None	74%
Cloud+	Insulation short of tee	Issues	None	69%



AS-BUILT ISOMETRIC — DOMESTIC WATER DROPS AT WALL PENETRATIONS

N.T.S. · DERIVED FROM PHOTO FT-04 · DIMENSIONS ESTIMATED

GENERAL NOTES

1. TYPE L COPPER, PRESS FITTINGS U.N.O.
2. PENETRATIONS: SLEEVED, FIRESTOPPED — SYSTEM NO. TBD
3. R5 OVERHEAD 2" DCW NOT SHOWN (OCCLUDED)

LEGEND

- DCW — Domestic Cold Water
- DHW — Domestic Hot Water
- CU — Copper (sugg.)
- Below 70% — verify

REVISION CLOUDS

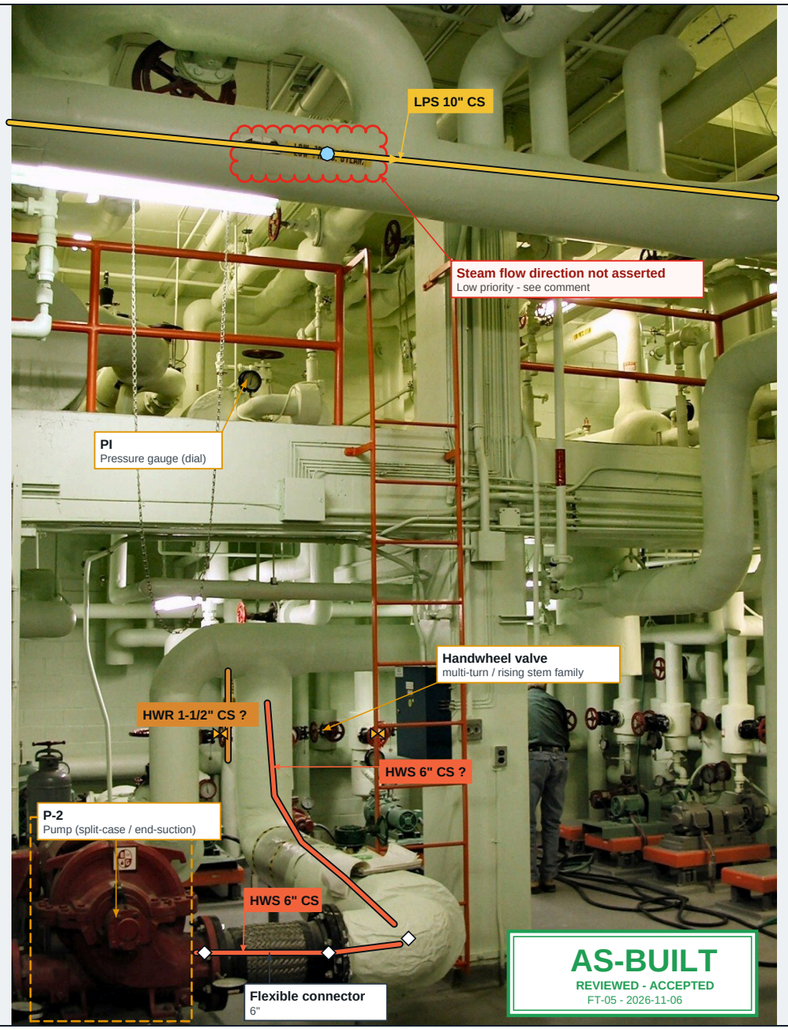
△1 Verify firestop system (Medium)
△2 Loose copper stub (High)
△3 Insulation short of tee (Low)

FIELD TRACE

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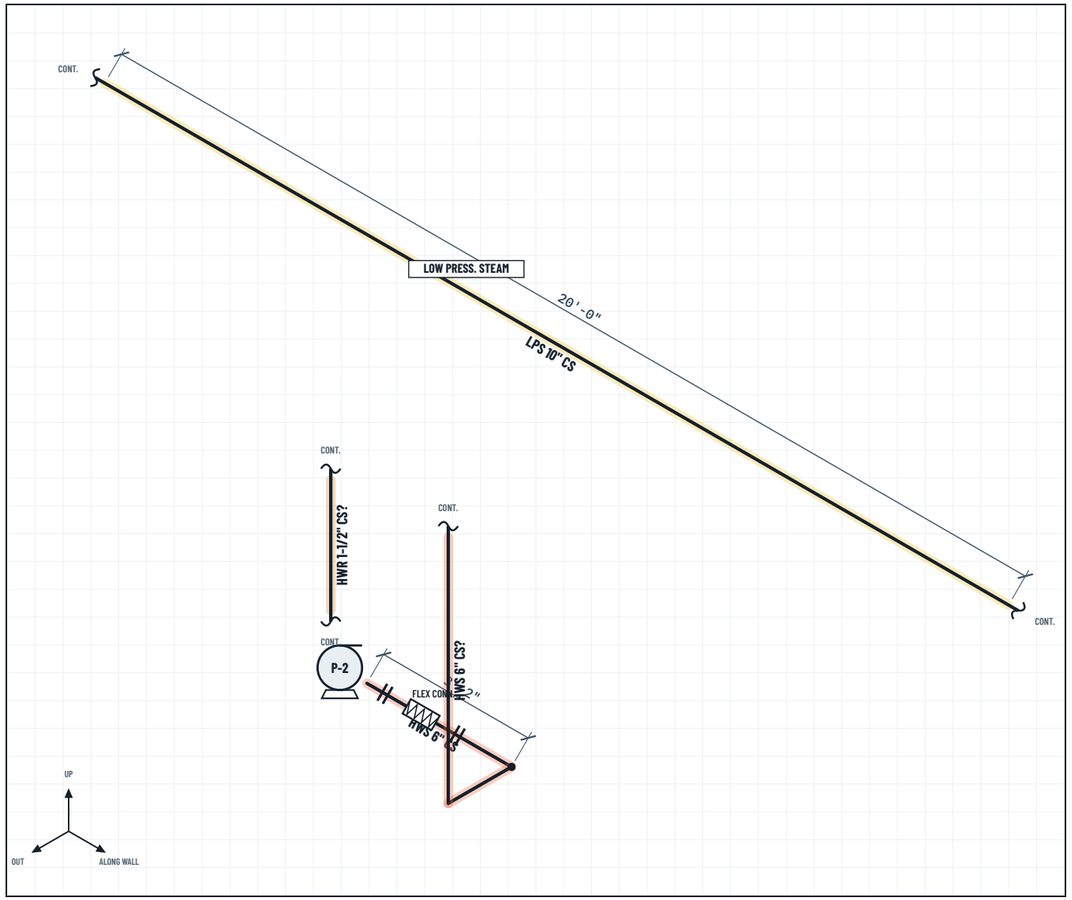
PROJECT: Sample Project — Mechanical Rough-In (DEMO-0142)			PREPARED BY: MIMETIC TECHNOLOGY	
DRAWING TITLE: AS-BUILT — DOMESTIC WATER DROPS AT WALL PENETRATIONS			LOCATION: Level 1 · Back-of-house wall W-12	
SOURCE PHOTO: FT-04 · 2026-11-10 15:31	DRAWN: FIELD TRACE AI	CHECKED: —	SCALE: N.T.S.	REV: A
STATUS: DRAFT — FOR REVIEW IN BLUEBEAM REVU 21.11			SHEET: FT-04	

SAMPLE DATA · PUBLIC LICENSED PHOTO · NOT A CLIENT JOB PHOTO · NOT FOR CONSTRUCTION



FIELD PHOTO FT-05 · CENTRAL PLANT · PUMP ROOM · 2026-11-06 10:20

MARKUPS = EDITABLE PDF ANNOTATIONS · AUTHOR: FIELD TRACE AI



AS-BUILT ISOMETRIC — LP STEAM HEADER & PUMP SET

N.T.S. · DERIVED FROM PHOTO FT-05 · DIMENSIONS ESTIMATED

MARKUP SUMMARY — 20 MARKUPS · 1 ISSUES · 2 BELOW 70%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Polyline	LPS 10" CS	Pipe Runs	Accepted	93%
Pipe Tag	LPS 10" CS	Labels	Accepted	70%
Polyline	HWS 6" CS	Pipe Runs	Accepted	87%
Pipe Tag	HWS 6" CS	Labels	Accepted	80%
Polyline	HWS 6" CS?	Pipe Runs	Accepted	74%
Pipe Tag	HWS 6" CS?	Labels	Accepted	55%
Polyline	HWR 1-1/2" CS?	Pipe Runs	Accepted	70%
Pipe Tag	HWR 1-1/2" CS?	Labels	Accepted	55%
Pump (split-case / end-suction)	P-2	Valves & Instruments	Accepted	88%
Flexible connector	6" Flexible connector	Fittings	Accepted	92%
Flange pair	6" Flange pair	Fittings	Accepted	90%
Flange pair	6" Flange pair	Fittings	Accepted	90%
90° elbow (insulated)	6" 90° elbow (insulated)	Fittings	Accepted	82%
Handwheel valve	2" Handwheel valve	Valves & Instruments	Accepted	78%
Handwheel valve	2" Handwheel valve	Valves & Instruments	Accepted	78%
Handwheel valve	2" Handwheel valve	Valves & Instruments	Accepted	76%
Pressure gauge (dial)	PI	Valves & Instruments	Accepted	83%
Pipe marker (read)	LOW PRESS. STEAM	Labels	Accepted	95%
Stamp	AS-BUILT — Reviewed	Labels	Accepted	100%
Cloud+	Steam flow direction not asserted	Issues	Accepted	90%

GENERAL NOTES

1. REVIEWED SET — ALL MARKUPS ACCEPTED
2. CONTINUATIONS SHOWN WITH BREAK SYMBOL
3. HEADER LENGTH IN FRAME ≈ 20'-0" (ESTIMATED)

LEGEND

- LPS — Low Pressure Steam
- HWS — Heating Hot Water Supply
- HWR — Heating Hot Water Return
- CS — Carbon steel (sugg.)
- Below 70% — verify

REVISION CLOUDS

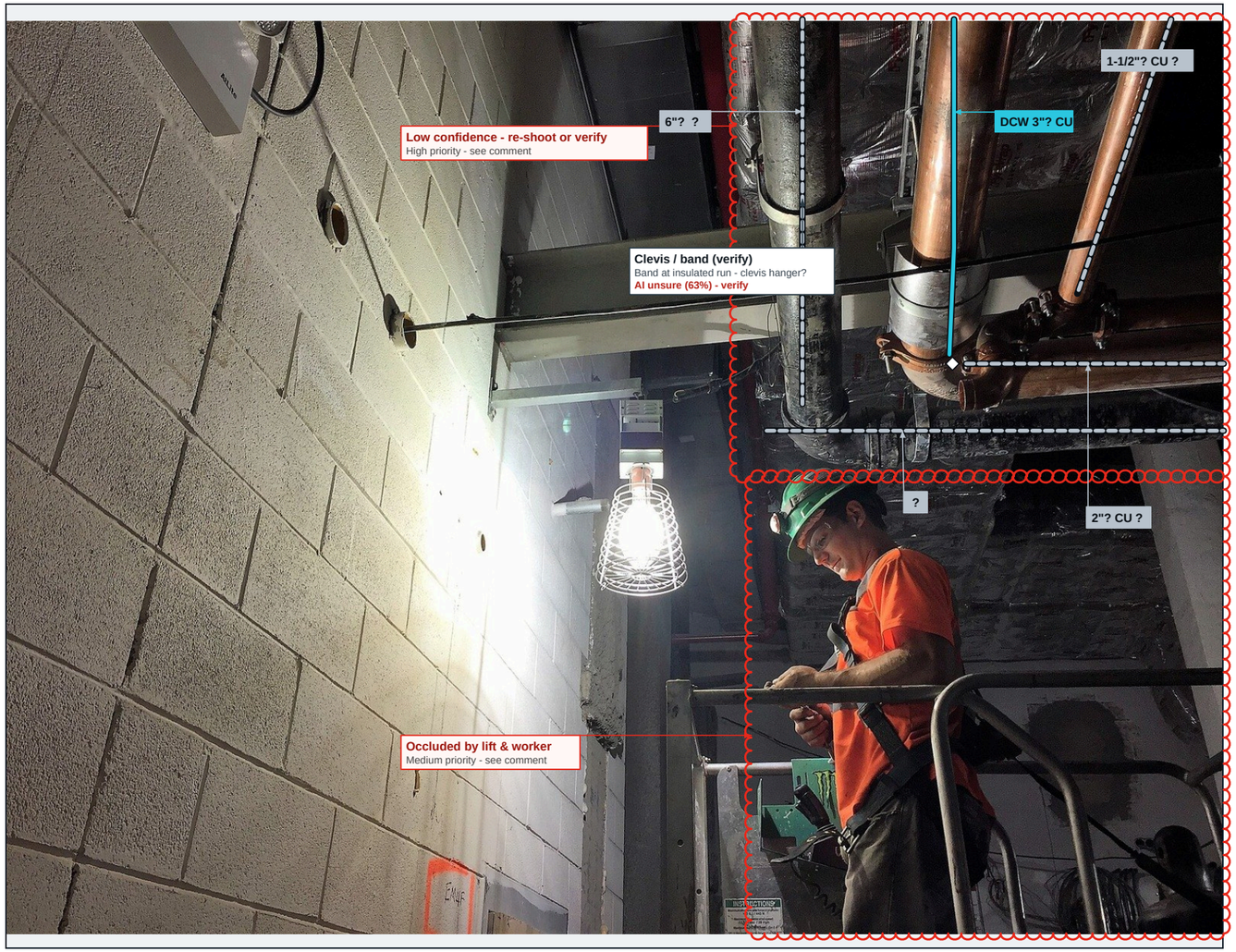
△1 Steam flow direction not asserted (Low)

FIELD TRACE

MIMETIC TECHNOLOGY · MIMETICTECH.AI
80/20 SOFTWARE · PROTOTYPE

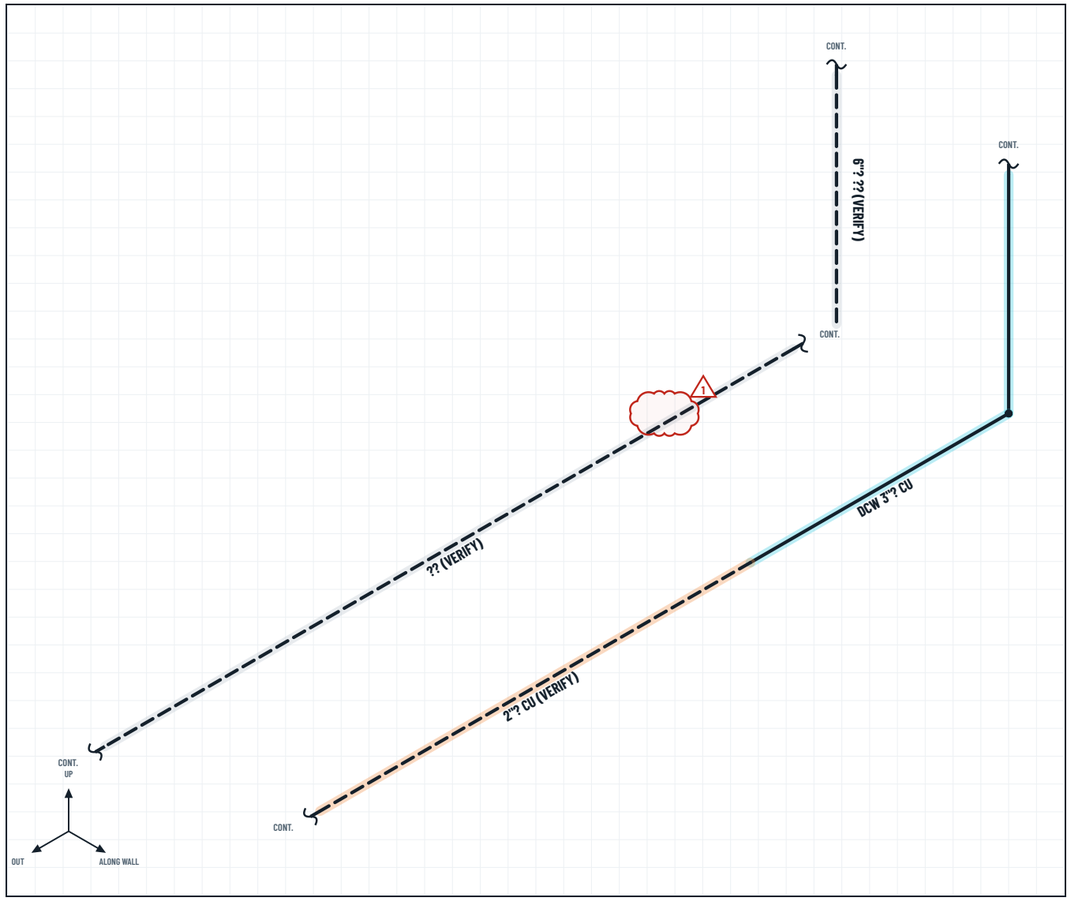
PROJECT: Sample Project — Mechanical Rough-In (DEMO-0142)			PREPARED BY: MIMETIC TECHNOLOGY	
DRAWING TITLE: AS-BUILT — LP STEAM HEADER & PUMP SET			LOCATION: Central plant · Pump room	
SOURCE PHOTO: FT-05 · 2026-11-06 10:20	DRAWN: FIELD TRACE AI	CHECKED: R. (REVIEWER)	SCALE: N.T.S.	REV: 0
STATUS: REVIEWED — ISSUED AS-BUILT			SHEET: FT-05	

SAMPLE DATA · PUBLIC LICENSED PHOTO · NOT A CLIENT JOB PHOTO · NOT FOR CONSTRUCTION



FIELD PHOTO FT-06 · LEVEL B1 · CORRIDOR D CEILING · 2026-11-11 16:48

MARKUPS = EDITABLE PDF ANNOTATIONS · AUTHOR: FIELD TRACE AI



AS-BUILT ISOMETRIC — CEILING SPACE ABOVE CORRIDOR — LOW LIGHT

N.T.S. · DERIVED FROM PHOTO FT-06 · DIMENSIONS ESTIMATED

MARKUP SUMMARY — 14 MARKUPS · 2 ISSUES · 10 BELOW 70%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Polyline	6"? ??	Pipe Runs	None	66%
Pipe Tag	6"? ??	Labels	None	50%
Polyline	DCW 3"? CU	Pipe Runs	None	71%
Pipe Tag	DCW 3"? CU	Labels	None	71%
Polyline	2"? CU	Pipe Runs	None	58%
Pipe Tag	2"? CU	Labels	None	58%
Polyline	1-1/2"? CU	Pipe Runs	None	61%
Pipe Tag	1-1/2"? CU	Labels	None	61%
Polyline	??	Pipe Runs	None	52%
Pipe Tag	??	Labels	None	50%
Clevis / band (verify)	Clevis / band (verify)	Hangers & Supports	None	63%
Elbow (copper)	3"? Elbow (copper)	Fittings	None	60%
Cloud+	**Low confidence — re-shoot or verify**	**Issues**	**None**	**95%**
Cloud+	**Occluded by lift & worker**	**Issues**	**None**	**90%**

GENERAL NOTES

1. NOT ISSUED — HELD FOR REVIEW
2. DASHED RUNS BELOW 70% CONFIDENCE
3. SIZES UNVERIFIED (NO SCALE REFERENCE)

LEGEND

- DCW — Domestic Cold Water
- CU — Copper (sugg.)
- ? — Not determinable (sugg.)
- Below 70% — verify

REVISION CLOUDS

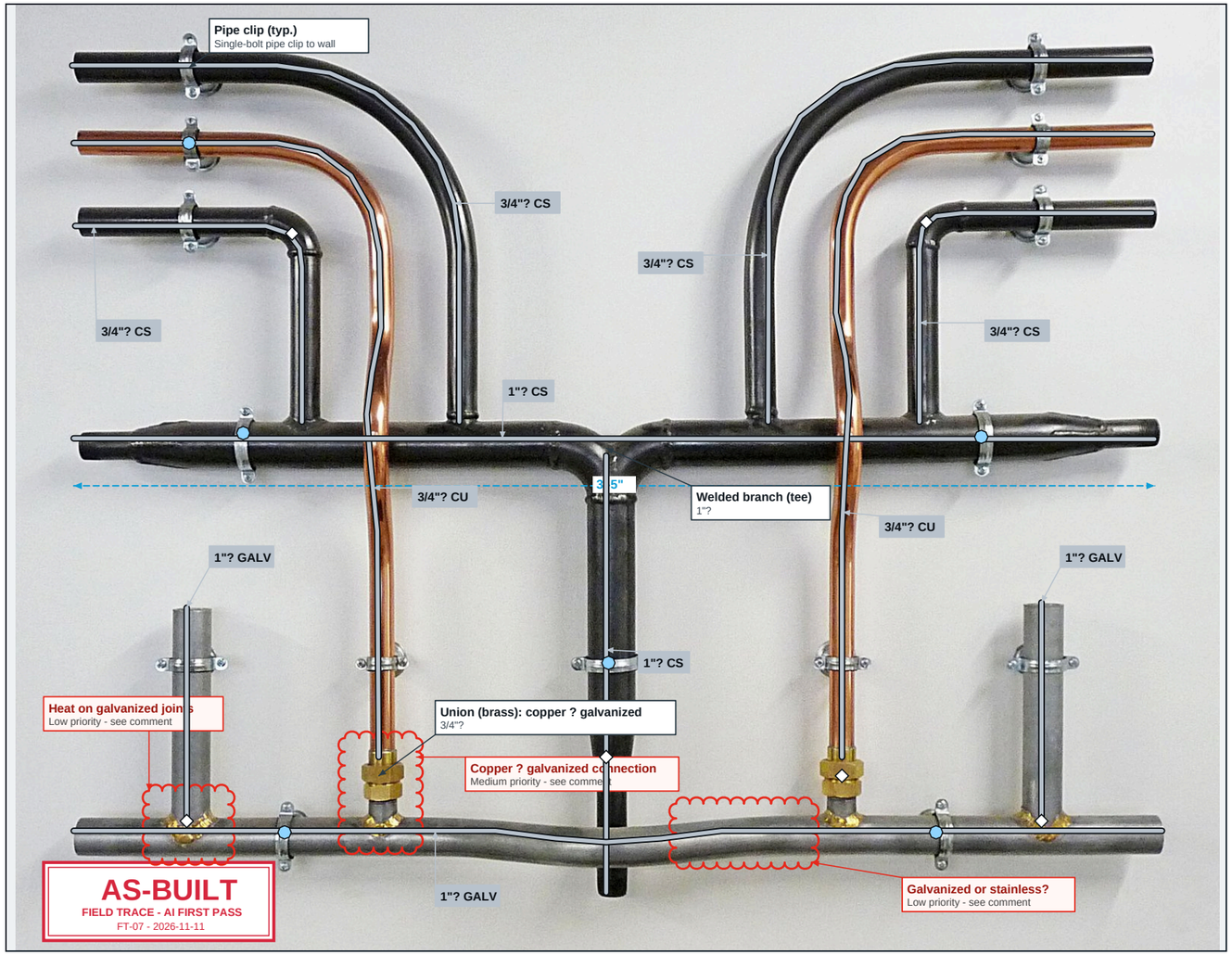
△1 Low confidence — re-shoot or verify (High)
△2 Occluded by lift & worker (Medium)

FIELD TRACE

MIMETIC TECHNOLOGY · MIMETICTECH.AI
80/20 SOFTWARE · PROTOTYPE

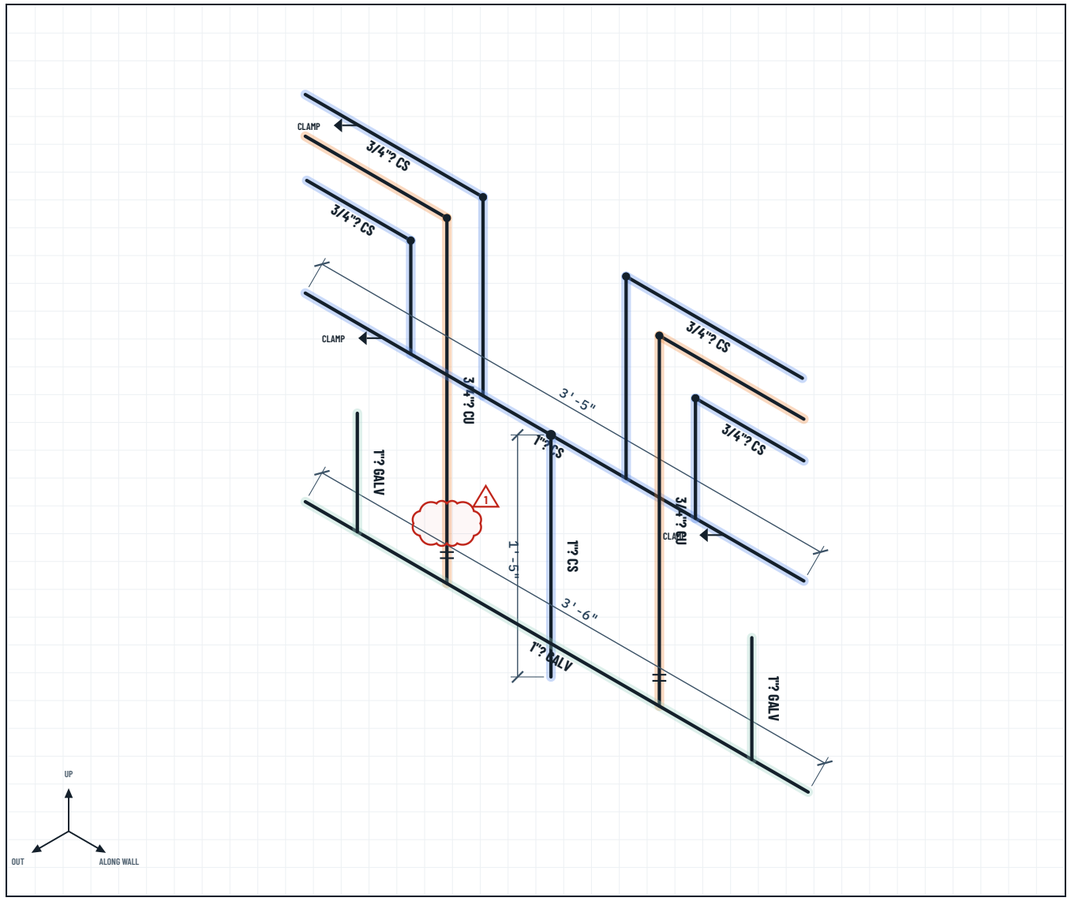
PROJECT: Sample Project — Mechanical Rough-In (DEMO-0142)			PREPARED BY: MIMETIC TECHNOLOGY	
DRAWING TITLE: AS-BUILT — CEILING SPACE ABOVE CORRIDOR — LOW LIGHT			LOCATION: Level B1 · Corridor D ceiling	
SOURCE PHOTO: FT-06 · 2026-11-11 16:48	DRAWN: FIELD TRACE AI	CHECKED: —	SCALE: N.T.S.	REV: A
STATUS: HELD — NEEDS FIELD VERIFICATION			SHEET: FT-06	

SAMPLE DATA · PUBLIC LICENSED PHOTO · NOT A CLIENT JOB PHOTO · NOT FOR CONSTRUCTION



FIELD PHOTO FT-07 · PREFAB SHOP · MOCK-UP WALL · 2026-11-11 08:10

MARKUPS = EDITABLE PDF ANNOTATIONS · AUTHOR: FIELD TRACE AI



AS-BUILT ISOMETRIC — MIXED MATERIALS — BLACK STEEL, COPPER, GALVANIZED

N.T.S. · DERIVED FROM PHOTO FT-07 · DIMENSIONS ESTIMATED

MARKUP SUMMARY — 42 MARKUPS · 3 ISSUES · 2 BELOW 70%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Polyline	3/4"? CS	Pipe Runs	None	90%
Pipe Tag	3/4"? CS	Labels	None	90%
Polyline	3/4"? CU	Pipe Runs	None	92%
Pipe Tag	3/4"? CU	Labels	None	92%
Polyline	3/4"? CS	Pipe Runs	None	90%
Pipe Tag	3/4"? CS	Labels	None	88%
Polyline	1"? CS	Pipe Runs	None	93%
Pipe Tag	1"? CS	Labels	None	90%
Polyline	3/4"? CS	Pipe Runs	None	90%
Pipe Tag	3/4"? CS	Labels	None	90%
Polyline	3/4"? CU	Pipe Runs	None	91%
Pipe Tag	3/4"? CU	Labels	None	91%
Polyline	3/4"? CS	Pipe Runs	None	90%
Pipe Tag	3/4"? CS	Labels	None	88%
Polyline	1"? CS	Pipe Runs	None	88%
Pipe Tag	1"? CS	Labels	None	88%
Polyline	1"? GALV	Pipe Runs	None	86%
Pipe Tag	1"? GALV	Labels	None	74%
Polyline	1"? GALV	Pipe Runs	None	88%
Pipe Tag	1"? GALV	Labels	None	72%
Polyline	1"? GALV	Pipe Runs	None	88%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Pipe Tag	1"? GALV	Labels	None	72%
Welded branch (tee)	1"? Welded branch (tee)	Fittings	None	88%
90° elbow (welded)	3/4"? 90° elbow (welded)	Fittings	None	86%
90° elbow (welded)	3/4"? 90° elbow (welded)	Fittings	None	86%
Concentric reducer (swaged)	Concentric reducer (swaged)	Fittings	None	80%
Union (brass): copper → galvanized	3/4"? Union (brass): copper → galvanized	Fittings	None	85%
Union (brass): copper → galvanized	3/4"? Union (brass): copper → galvanized	Fittings	None	85%
Brazed branch joint	Brazed branch joint	Fittings	None	78%
Brazed branch joint	Brazed branch joint	Fittings	None	78%
Pipe clip (typ.)	Pipe clip (typ.)	Hangers & Supports	None	90%
Pipe clip	Pipe clip	Hangers & Supports	None	88%
Pipe clip	Pipe clip	Hangers & Supports	None	88%
Pipe clip	Pipe clip	Hangers & Supports	None	88%
Pipe clip	Pipe clip	Hangers & Supports	None	88%
Pipe clip	Pipe clip	Hangers & Supports	None	86%
Pipe clip	Pipe clip	Hangers & Supports	None	86%
Dimension	3'-5"	Dimensions	None	66%
Stamp	AS-BUILT — Draft	Labels	None	100%
Cloud+	Copper ↔ galvanized connection	Issues	None	82%
Cloud+	Heat on galvanized joints	Issues	None	70%
Cloud+	Galvanized or stainless?	Issues	None	66%

GENERAL NOTES

1. SCALE FROM COPPER OD (22 MM ASSUMED): LENGTHS APPROXIMATE
2. BENDS SHOWN AS CORNERS. BENT PIPE, NO FITTING
3. MATERIALS SUGGESTED: CS BLACK · CU · GALV. VERIFY

LEGEND

- CS — Carbon steel (sugg.)
- CU — Copper (sugg.)
- GALV — Galvanized steel (sugg.)
- Below 70% — verify

REVISION CLOUDS

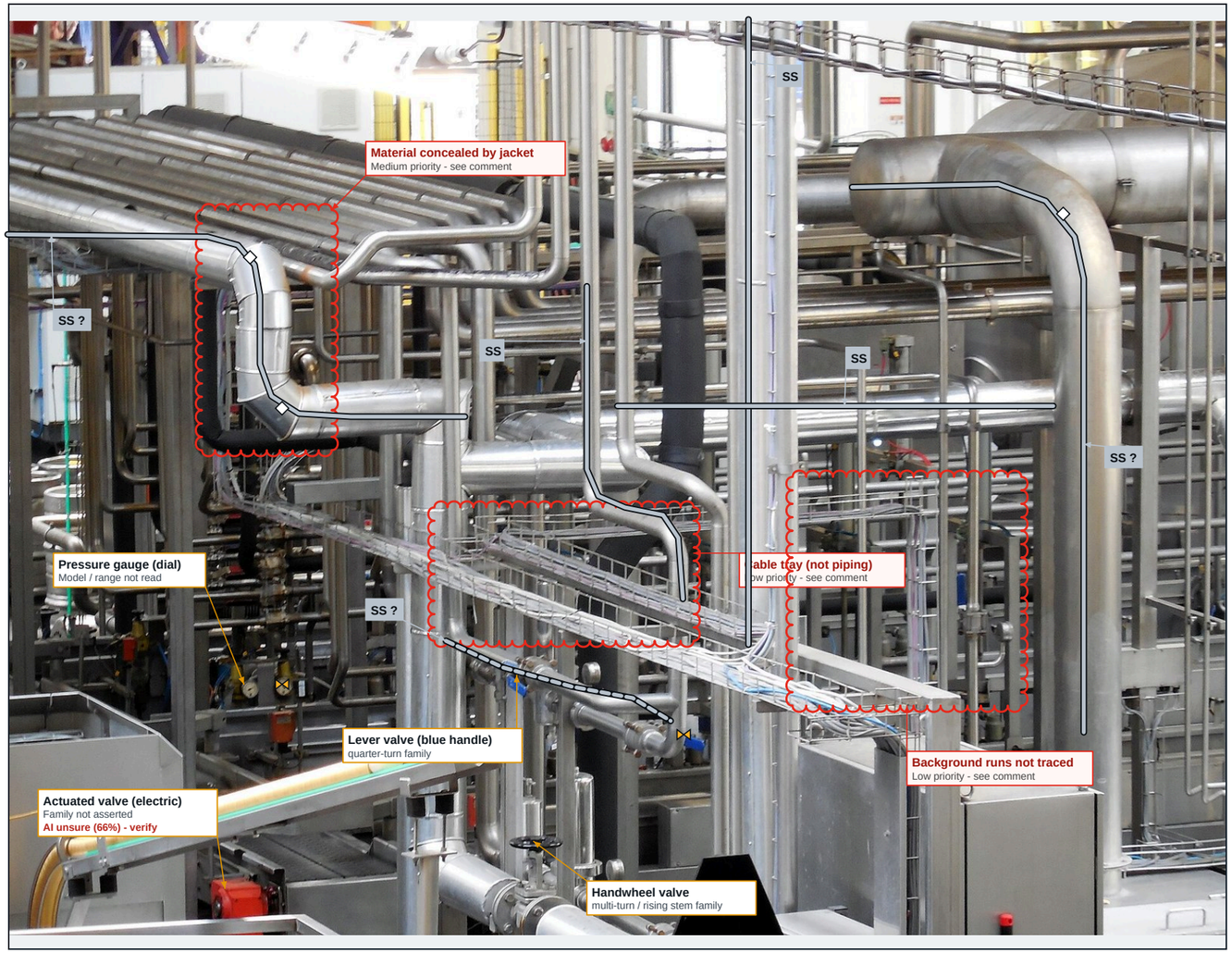
△1 Copper ↔ galvanized connection (Medium)
△2 Heat on galvanized joints (Low)
△3 Galvanized or stainless? (Low)

FIELD TRACE

MIMETIC TECHNOLOGY · MIMETICTECH.AI
80/20 SOFTWARE · PROTOTYPE

PROJECT: Sample Project — Mechanical Rough-In (DEMO-0142)			PREPARED BY: MIMETIC TECHNOLOGY
DRAWING TITLE: AS-BUILT — MIXED MATERIALS — BLACK STEEL, COPPER, GALVANIZED			LOCATION: Prefab shop · mock-up wall
SOURCE PHOTO: FT-07 · 2026-11-11 08:10	DRAWN: FIELD TRACE AI	CHECKED: —	SCALE: N.T.S. / REV: A
STATUS: DRAFT — FOR REVIEW IN BLUEBEAM REVU 21.11			SHEET: FT-07

SAMPLE DATA · PUBLIC LICENSED PHOTO · NOT A CLIENT JOB PHOTO · NOT FOR CONSTRUCTION



FIELD PHOTO FT-08 · PROCESS AREA · UTILITY RACK · 2026-11-11 10:05

MARKUPS = EDITABLE PDF ANNOTATIONS · AUTHOR: FIELD TRACE AI

MARKUP SUMMARY — 24 MARKUPS · 3 ISSUES · 6 BELOW 70%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Polyline	SS?	Pipe Runs	None	84%
Pipe Tag	SS?	Labels	None	50%
Polyline	SS?	Pipe Runs	None	82%
Pipe Tag	SS?	Labels	None	50%
Polyline	SS	Pipe Runs	None	80%
Pipe Tag	SS	Labels	None	80%
Polyline	SS	Pipe Runs	None	70%
Pipe Tag	SS	Labels	None	70%
Polyline	SS	Pipe Runs	None	74%
Pipe Tag	SS	Labels	None	74%
Polyline	SS?	Pipe Runs	None	62%
Pipe Tag	SS?	Labels	None	62%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Lever valve (blue handle)	Lever valve (blue handle)	Valves & Instruments	None	74%
Lever valve (blue handle)	Lever valve (blue handle)	Valves & Instruments	None	72%
Handwheel valve	Handwheel valve	Valves & Instruments	None	70%
Actuated valve (electric)	Actuated valve (electric)	Valves & Instruments	None	66%
Pressure gauge (dial)	Pressure gauge (dial)	Valves & Instruments	None	70%
Pressure gauge (dial)	Pressure gauge (dial)	Valves & Instruments	None	66%
Elbow (jacketed)	Elbow (jacketed)	Fittings	None	84%
Elbow (jacketed)	Elbow (jacketed)	Fittings	None	82%
Elbow (jacketed)	Elbow (jacketed)	Fittings	None	82%
Cloud+	Material concealed by jacket	Issues	None	80%
Cloud+	Cable tray (not piping)	Issues	None	75%
Cloud+	Background runs not traced	Issues	None	70%

AS-BUILT ISOMETRIC: HELD

Not drafted: dense rack, no scale reference. Confirm one bare-pipe size in the workspace and lengths follow.

Run	Material (sugg.)	Size	OD	Schedule (sugg.)
R1	SS ?	?	Jacket OD only	—
R2	SS ?	?	Jacket OD only	—
R3	SS	?	NPS (S) or OD tube convention · needs a scale	Sch 10S
R4	SS	?	NPS (S) or OD tube convention · needs a scale	Sch 10S
R5	SS	?	NPS (S) or OD tube convention · needs a scale	Sch 10S
R6	SS ?	?	NPS (S) or OD tube convention · needs a scale	—

Wall thickness is not visible from a photo. Rating and corrosion allowance need verification.

RUN SCHEDULE — PROCESS RACK — STAINLESS AND JACKETED LINES

NO ISO DRAFTED · MATERIALS AND SCHEDULES SUGGESTED

GENERAL NOTES

1. Not drafted: dense rack, no scale reference. Confirm one bare-pipe size in the workspace and lengths follow.
2. SCHEDULES ARE SUGGESTED. WALL IS NOT VISIBLE IN A PHOTO. VERIFY

LEGEND

SS — Stainless steel (sugg.)

Below 70% — verify

REVISION CLOUDS

△1 Material concealed by jacket (Medium)

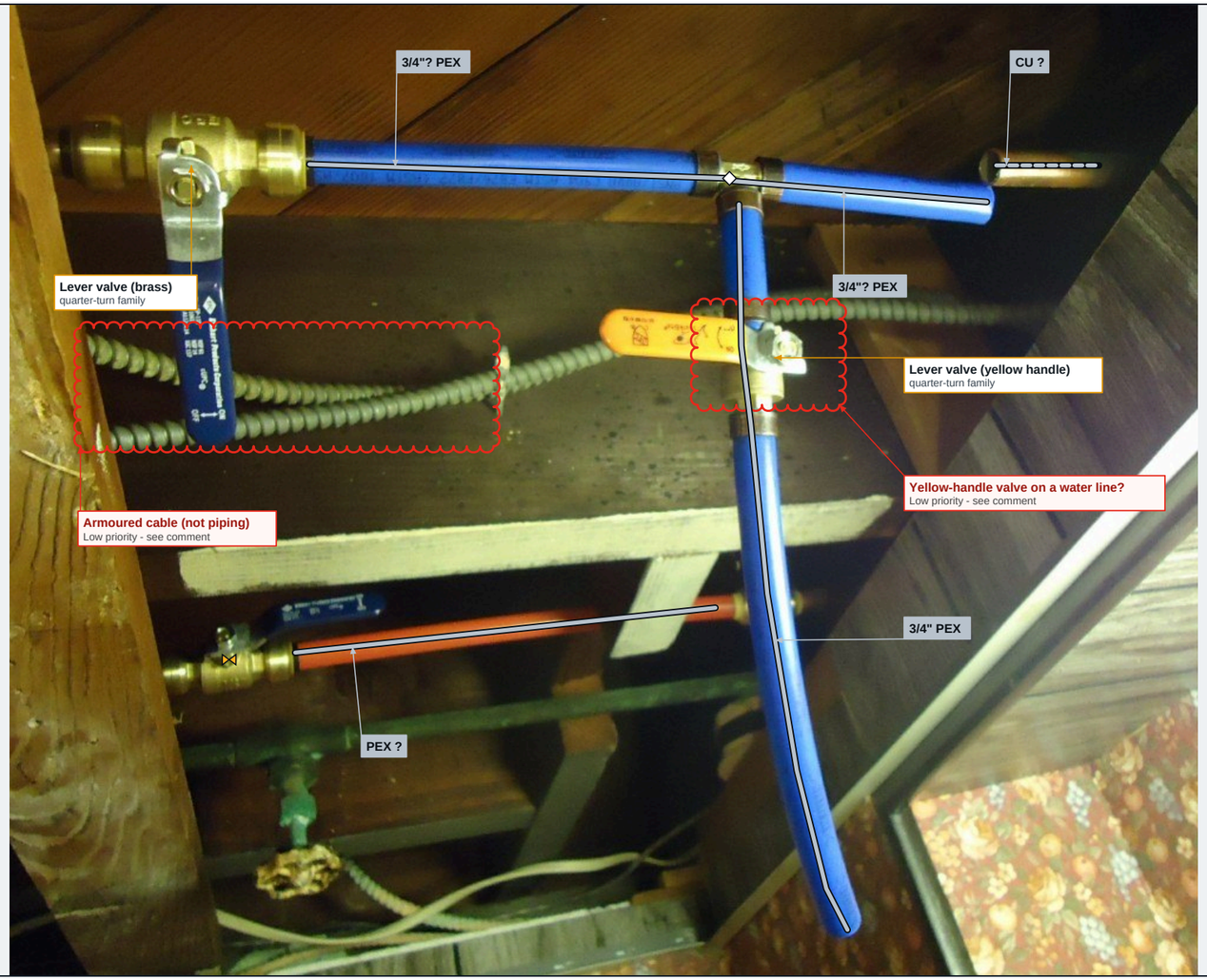
△2 Cable tray (not piping) (Low)

△3 Background runs not traced (Low)

FIELD TRACE

MIMETIC TECHNOLOGY · MIMETICTECH.AI
80/20 SOFTWARE · PROTOTYPE

PROJECT: Sample Project — Mechanical Rough-In (DEMO-0142)			PREPARED BY: MIMETIC TECHNOLOGY	
DRAWING TITLE: AS-BUILT — PROCESS RACK — STAINLESS AND JACKETED LINES			LOCATION: Process area · utility rack	
SOURCE PHOTO: FT-08 · 2026-11-11 10:05	DRAWN: FIELD TRACE AI	CHECKED: —	SCALE: N.T.S.	REV: A
STATUS: DRAFT — FOR REVIEW IN BLUEBEAM REVU 21.11			SHEET: FT-08	



FIELD PHOTO FT-09 · RESIDENTIAL · BASEMENT CEILING · 2026-11-11 14:22

MARKUPS = EDITABLE PDF ANNOTATIONS · AUTHOR: FIELD TRACE AI

AS-BUILT ISOMETRIC: HELD

Not drafted as an iso. Lengths come from the assumed 3/4" PEX scale.

Run	Material (sugg.)	Size	OD	Schedule (sugg.)
R1	PEX	3/4"?	0.875"?	SDR 9
R2	PEX	3/4"?	0.875"?	SDR 9
R3	CU ?	?	CTS convention · needs a scale	—
R4	PEX	3/4"	0.875"	SDR 9
R5	PEX ?	?	CTS convention · needs a scale	SDR 9

Wall thickness is not visible from a photo. Rating and corrosion allowance need verification.

RUN SCHEDULE — PEX AND COPPER — BASEMENT CEILING

NO ISO DRAFTED · MATERIALS AND SCHEDULES SUGGESTED

MARKUP SUMMARY — 16 MARKUPS · 2 ISSUES · 4 BELOW 70%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Polyline	3/4"? PEX	Pipe Runs	None	86%
Pipe Tag	3/4"? PEX	Labels	None	78%
Polyline	3/4"? PEX	Pipe Runs	None	82%
Pipe Tag	3/4"? PEX	Labels	None	76%
Polyline	CU?	Pipe Runs	None	60%
Pipe Tag	CU?	Labels	None	55%
Polyline	3/4" PEX	Pipe Runs	None	85%
Pipe Tag	3/4" PEX	Labels	None	80%
Polyline	PEX?	Pipe Runs	None	72%
Pipe Tag	PEX?	Labels	None	64%
Lever valve (brass)	Lever valve (brass)	Valves & Instruments	None	84%
Lever valve (yellow handle)	Lever valve (yellow handle)	Valves & Instruments	None	80%
Lever valve (brass)	Lever valve (brass)	Valves & Instruments	None	70%
Tee (brass, crimp)	3/4"? Tee (brass, crimp)	Fittings	None	84%
Cloud+	Armoured cable (not piping)	Issues	None	80%
Cloud+	Yellow-handle valve on a water line?	Issues	None	60%

GENERAL NOTES

1. Not drafted as an iso. Lengths come from the assumed 3/4" PEX scale.
2. SCHEDULES ARE SUGGESTED. WALL IS NOT VISIBLE IN A PHOTO. VERIFY

LEGEND

- CU — Copper (sugg.)
- COMP — Composite & plastic (sugg.)
- Below 70% — verify

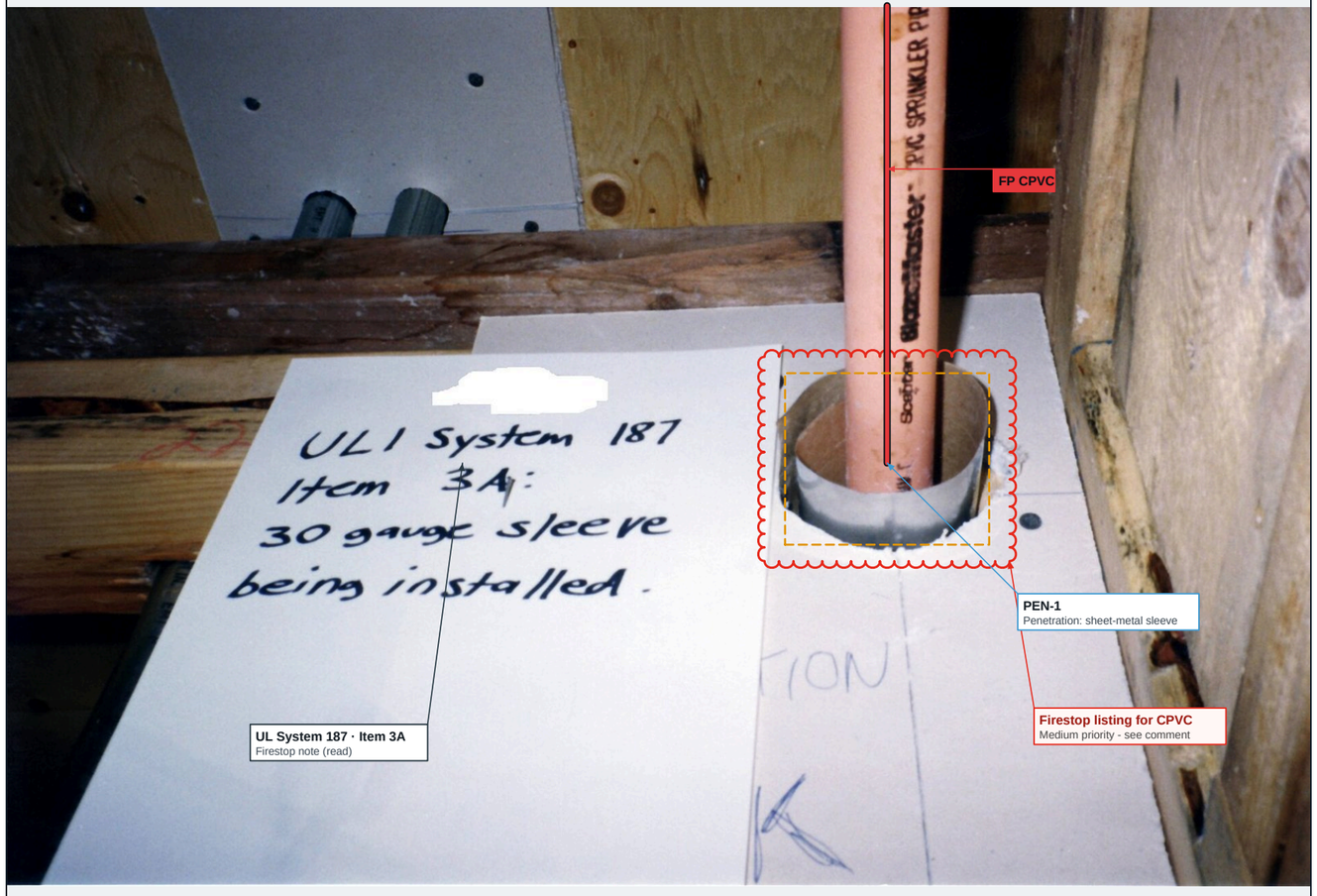
REVISION CLOUDS

△1 Armoured cable (not piping) (Low)
△2 Yellow-handle valve on a water line? (Low)

FIELD TRACE

MIMETIC TECHNOLOGY · MIMETICTECH.AI
80/20 SOFTWARE · PROTOTYPE

PROJECT: Sample Project — Mechanical Rough-In (DEMO-0142)			PREPARED BY: MIMETIC TECHNOLOGY	
DRAWING TITLE: AS-BUILT — PEX AND COPPER — BASEMENT CEILING			LOCATION: Residential · basement ceiling	
SOURCE PHOTO: FT-09 · 2026-11-11 14:22	DRAWN: FIELD TRACE AI	CHECKED: —	SCALE: N.T.S.	REV: A
STATUS: DRAFT — FOR REVIEW IN BLUEBEAM REVU 21.11			SHEET: FT-09	



FIELD PHOTO FT-10 · WOOD-FRAME FLOOR · PENETRATION · 2026-11-11 15:40

MARKUPS = EDITABLE PDF ANNOTATIONS · AUTHOR: FIELD TRACE AI

MARKUP SUMMARY — 5 MARKUPS · 1 ISSUES · 0 BELOW 70%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Polyline	FP CPVC	Pipe Runs	None	90%
Pipe Tag	FP CPVC	Labels	None	90%
Penetration: sheet-metal sleeve	PEN-1	Hangers & Supports	None	88%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Firestop note (read)	UL System 187 · Item 3A	Labels	None	80%
Cloud+	Firestop listing for CPVC	Issues	None	80%

AS-BUILT ISOMETRIC: HELD

Single vertical run, no scale. Not drafted.

Run	Material (sugg.)	Size	OD	Schedule (sugg.)
R1	CPVC	?	IPS convention · needs a scale	SDR 13.5

Wall thickness is not visible from a photo. Rating and corrosion allowance need verification.

RUN SCHEDULE — CPVC SPRINKLER PIPE AT A FIRESTOP SLEEVE

NO ISO DRAFTED · MATERIALS AND SCHEDULES SUGGESTED

GENERAL NOTES

1. Single vertical run, no scale. Not drafted.
2. SCHEDULES ARE SUGGESTED. WALL IS NOT VISIBLE IN A PHOTO. VERIFY

LEGEND

- FP — Fire Protection
- COMP — Composite & plastic (sugg.)
- Below 70% — verify

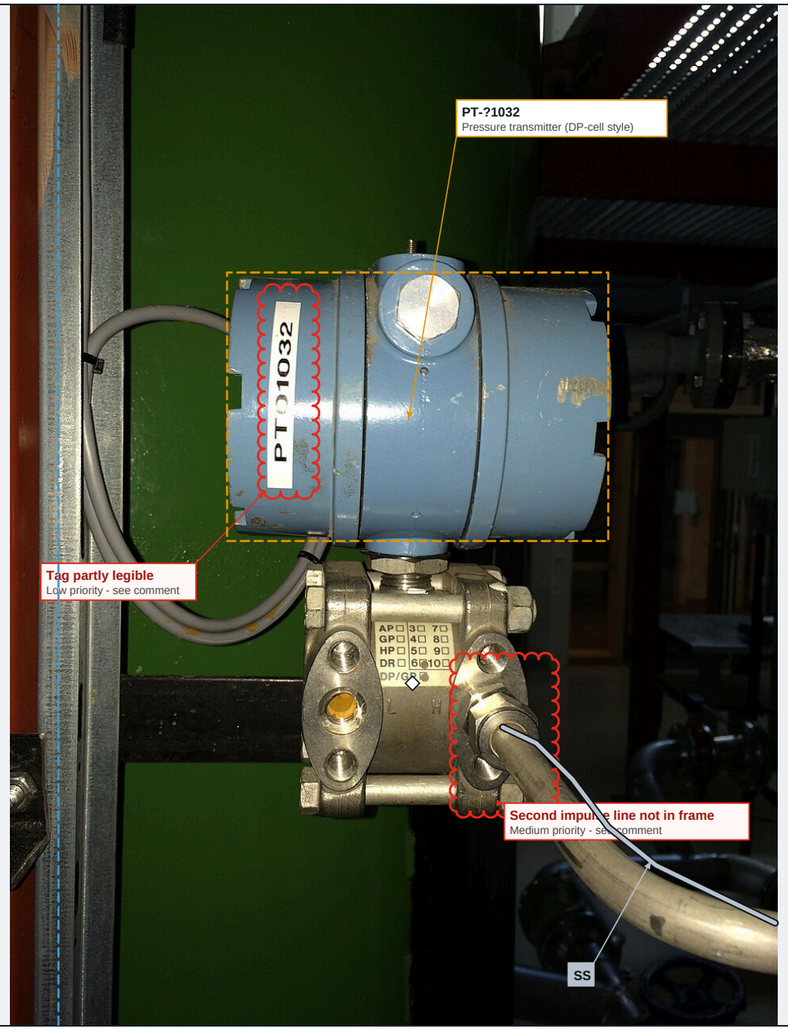
REVISION CLOUDS

△1 Firestop listing for CPVC (Medium)

FIELD TRACE

MIMETIC TECHNOLOGY · MIMETICTECH.AI
80/20 SOFTWARE · PROTOTYPE

PROJECT: Sample Project — Mechanical Rough-In (DEMO-0142)			PREPARED BY: MIMETIC TECHNOLOGY	
DRAWING TITLE: AS-BUILT — CPVC SPRINKLER PIPE AT A FIRESTOP SLEEVE			LOCATION: Wood-frame floor · penetration	
SOURCE PHOTO: FT-10 · 2026-11-11 15:40	DRAWN: FIELD TRACE AI	CHECKED: —	SCALE: N.T.S.	REV: A
STATUS: DRAFT — FOR REVIEW IN BLUEBEAM REVU 21.11			SHEET: FT-10	



FIELD PHOTO FT-11 · INSTRUMENT STAND · 2026-11-11 16:02

MARKUPS = EDITABLE PDF ANNOTATIONS · AUTHOR: FIELD TRACE AI

MARKUP SUMMARY — 7 MARKUPS · 2 ISSUES · 1 BELOW 70%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Polyline	SS	Pipe Runs	None	82%
Pipe Tag	SS	Labels	None	74%
Pressure transmitter (DP-cell style)	PT-?1032	Valves & Instruments	None	84%
Process flange (sensor module)	Process flange (sensor module)	Fittings	None	80%

SUBJECT	LABEL	LAYER	STATUS	CONF.
Strut channel (stand)	Strut channel (stand)	Hangers & Supports	None	80%
Cloud+	Second impulse line not in frame	Issues	None	66%
Cloud+	Tag partly legible	Issues	None	80%

AS-BUILT ISOMETRIC: HELD

Instrument hook-up. Not drafted as an iso.

Run	Material (sugg.)	Size	OD	Schedule (sugg.)
R1	SS	?	Tube OD convention · needs a scale	—

Wall thickness is not visible from a photo. Rating and corrosion allowance need verification.

RUN SCHEDULE — PRESSURE TRANSMITTER ON STAINLESS IMPULSE TUBING

NO ISO DRAFTED · MATERIALS AND SCHEDULES SUGGESTED

GENERAL NOTES

1. Instrument hook-up. Not drafted as an iso.
2. SCHEDULES ARE SUGGESTED. WALL IS NOT VISIBLE IN A PHOTO. VERIFY

LEGEND

SS — Stainless steel (sugg.)

Below 70% — verify

REVISION CLOUDS

△1 Second impulse line not in frame (Medium)

△2 Tag partly legible (Low)

FIELD TRACE

MIMETIC TECHNOLOGY · MIMETICTECH.AI
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PROJECT: Sample Project — Mechanical Rough-In (DEMO-0142)			PREPARED BY: MIMETIC TECHNOLOGY	
DRAWING TITLE: AS-BUILT — PRESSURE TRANSMITTER ON STAINLESS IMPULSE TUBING			LOCATION: Instrument stand	
SOURCE PHOTO: FT-11 · 2026-11-11 16:02	DRAWN: FIELD TRACE AI	CHECKED: —	SCALE: N.T.S.	REV: A
STATUS: DRAFT — FOR REVIEW IN BLUEBEAM REVU 21.11			SHEET: FT-11	

SAMPLE DATA · PUBLIC LICENSED PHOTO · NOT A CLIENT JOB PHOTO · NOT FOR CONSTRUCTION

PIPE — BY MATERIAL / SYSTEM / SIZE

MATERIAL (SUGG.)	SYSTEM	SIZE	OD	SCHED. (SUGG.)	LF	PHOTOS	
CS	CHWR	8"	8.625" (if 8" NPS)	Std (Sch 40)	30.0	FT-03	VERIFY
CS	CHWS	8"	8.625" (if 8" NPS)	Std (Sch 40)	30.0	FT-03	VERIFY
CS	CTR	6"	6.625"	Sch 40	5.9	FT-01	
CS	CTS	6"	6.625"	Sch 40	9.9	FT-01	
CS	CWR	6"	6.625"	Sch 40	5.9	FT-01	
CS	CWS	6"	6.625"	Sch 40	9.6	FT-01	
CS	FP	4"	4.5"	Sch 10	48.1	FT-03	
CS	HWR	1-1/2"	—	Sch 40	2.5	FT-05	VERIFY
CS	HWS	4"	4.5" (if 4" NPS)	Std (Sch 40)	31.6	FT-03	VERIFY
CS	HWS	6"	6.625"	Sch 40	9.7	FT-05	VERIFY
CS	HWS	2-1/2"	2.875"	Sch 10	7.8	FT-02	
CS	LPS	10"	10.75" (if 10" NPS)	Sch 40	20.0	FT-05	
CS	UNK	3/4"?	≈1.05"	Sch 40	7.7	FT-07	
CS	UNK	1"?	≈1.32"	Sch 40	4.9	FT-07	
SS	UNK	?	Jacket OD only	—	—*	FT-08 FT-11	VERIFY
CU	DCW	3"?	3.125"?	Type L	5.4	FT-06	
CU	DCW	1-1/4"	—	Type L	5.0	FT-04	
CU	DCW	3/4"	0.875"	Type L	2.4	FT-04	
CU	DCW	2"	2.125"	Type L	—*	FT-04	
CU	DHW	1"	—	Type L	4.4	FT-04	
CU	DHW	3/4"	0.875"	Type L	2.9	FT-04	
CU	UNK	3/4"?	≈0.875" (22 mm)	Type L	6.4	FT-07	
CU	UNK	2"?	2.125"?	Type L	5.0	FT-06	VERIFY
CU	UNK	3/4"	0.875"	Type L	1.0	FT-04	VERIFY
CU	UNK	?	—	—	0.2	FT-09	VERIFY
CU	UNK	1-1/2"?	1.625"?	Type L	—*	FT-06	VERIFY
GALV	HWS	2"	2.375"	Sch 40	3.5	FT-02	VERIFY
GALV	UNK	1"?	≈1.32"	Sch 40	4.9	FT-07	
CPVC	FP	?	—	SDR 13.5	—*	FT-10	
PEX	UNK	3/4"	0.875"	SDR 9	1.7	FT-09	
PEX	UNK	3/4"?	0.875"?	SDR 9	1.6	FT-09	
PEX	UNK	?	—	SDR 9	1.0	FT-09	VERIFY
?	UNK	?	—	—	8.0	FT-06	VERIFY
?	UNK	3/4"?	—	—	3.7	FT-02	VERIFY
?	UNK	6"?	—	—	2.5	FT-06	VERIFY
Total measured					**283.3**		
Carbon steel					223.7		
Stainless steel					0.0	7 run(s) not measured	
Copper					32.7	2 run(s) not measured	
Galvanized steel					8.4		
Composite & plastic					4.2	1 run(s) not measured	
Not determinable					14.2		

* includes runs on photos with no scale (not measured, not guessed). Schedules are suggestions: the wall is not visible in a photo, so pressure / temperature rating and remaining corrosion allowance must be verified.

OPEN ISSUES

PHOTO	SEVERITY	ISSUE	COMMENT
FT-01	Medium	**No support visible at CWR offset**	6" CWR 45° offset between valve and HX shows no hanger or floor stand in frame. Verify suppo…
FT-01	Low	**Nameplate not legible**	PHE-1 tag assumed from drawing schedule. Capture nameplate close-up (model / serial) for O&M.
FT-02	High	**Open tee outlet**	Branch outlet on 2-1/2" tee is open. Cap or connect before hydrostatic test.
FT-02	Medium	**Material transition at coupling**	Finish changes at coupling (black → galvanized?). If dissimilar metals, verify dielectric/tr…
FT-03	Medium	**Fitting insulation incomplete**	Insulation at tee cluster is irregular / covers appear missing. Complete fitting covers befo…
FT-03	Low	**Clearance at FP crossing**	Upper FP main crosses tight above CHWS. Verify clearance and access to FP fittings.
FT-04	Medium	**Verify firestop system**	Sealant present at 3 sleeves. Confirm listed firestop system number and label each penetrati…
FT-04	High	**Loose copper stub**	3/4" copper stub at PEN-3 is not connected or capped. Confirm intent; cap and test or remove.
FT-04	Low	**Insulation short of tee**	Insulation stops short at DCW tee. Complete before inspection.
FT-05	Low	**Steam flow direction not asserted**	Marker arrow is double-headed. Flow direction left unassigned; add from P&ID.
FT-06	High	**Low confidence — re-shoot or verify**	Lamp glare and shadow: 3 of 5 runs below 70% confidence. Not auto-drafted. Re-shoot with fla…
FT-06	Medium	**Occluded by lift & worker**	Lower ceiling and branch continuations hidden. Second angle needed before as-built.
FT-07	Medium	**Copper ↔ galvanized connection**	Copper joins galvanized through a brass union. Brass is a common transition, but confirm the…
FT-07	Low	**Heat on galvanized joints**	Brazing burns the zinc off near the joint. Confirm the joint method is allowed and touch up …
FT-07	Low	**Galvanized or stainless?**	Matte zinc-grey suggests galvanized. Weathered stainless can look similar. A magnet or spang…
FT-08	Medium	**Material concealed by jacket**	Aluminium jacket over insulation. Pipe material and OD can't be seen, so SS is a suggestion …
FT-08	Low	**Cable tray (not piping)**	Wire tray excluded from the trace. Check clearance where it crosses the stainless lines.
FT-08	Low	**Background runs not traced**	Dense rack. Lines behind the front layer need a second angle before an as-built.
FT-09	Low	**Armoured cable (not piping)**	Flexible armoured electrical cable excluded from the trace, so it stays out of the takeoff.
FT-09	Low	**Yellow-handle valve on a water line?**	Yellow levers often mean gas. Confirm the service and the valve's rating.
FT-10	Medium	**Firestop listing for CPVC**	Note cites UL System 187, Item 3A. Confirm that listed system covers CPVC at this size. Plas…
FT-11	Medium	**Second impulse line not in frame**	DP-style transmitters usually have high- and low-side connections. Only one tube is visible.…
FT-11	Low	**Tag partly legible**	One character of the tag label isn't readable. Confirm against the instrument index.

VALVES, INSTRUMENTS, FITTINGS & SUPPORTS

CATEGORY	ITEM	SIZE	QTY	PHOTOS	
Valves	Actuated valve (electric) · Electric actuator (red housing) · family not asserted	—	1	FT-08	VERIFY
Valves	Gear-operated valve · Gear operator · quarter-turn (ball or butterfly)	6"	1	FT-01	
Valves	Handwheel valve · Handwheel · multi-turn / rising stem (gate, globe or plug)	6"	2	FT-01	
Valves	Handwheel valve · Handwheel · multi-turn / rising stem (gate, globe or plug)	2"	3	FT-05	
Valves	Handwheel valve · Handwheel · multi-turn / rising stem (gate, globe or plug)	—	1	FT-08	
Valves	Lever valve (blue handle) · Lever · quarter-turn (ball or butterfly)	—	2	FT-08	
Valves	Lever valve (brass) · Lever · quarter-turn (ball or butterfly)	—	2	FT-09	
Valves	Lever valve (drain / vent) · Lever · quarter-turn (ball or butterfly)	3/4"	2	FT-01	
Valves	Lever valve (yellow handle) · Lever · quarter-turn (ball or butterfly)	—	1	FT-09	
Valves	Sprinkler branch outlet	1"	1	FT-03	
Instruments	Pressure gauge (dial) · Pressure · model / range not read	—	3	FT-05 FT-08	VERIFY
Instruments	Pressure transmitter (DP-cell style) · Pressure · model / range not read	—	1	FT-11	
Instruments	Thermometer in thermowell · Temperature · model / range not read	—	4	FT-01	
Equipment	Plate & frame heat exchanger	—	1	FT-01	
Equipment	Pump (split-case / end-suction)	—	1	FT-05	
Fittings	45° elbow	6"	2	FT-01	
Fittings	90° elbow (grooved)	2-1/2"	2	FT-02	
Fittings	90° elbow (insulated)	6"	1	FT-05	
Fittings	90° elbow (welded)	3/4"?	2	FT-07	
Fittings	90° LR elbow	6"	2	FT-01	
Fittings	Brazed branch joint	—	2	FT-07	
Fittings	Cap (stub-out)	1"	1	FT-04	
Fittings	Cap (stub-out)	1-1/4"	1	FT-04	
Fittings	Concentric reducer (swaged)	—	1	FT-07	
Fittings	Coupling / transition	2-1/2" × 2"	1	FT-02	
Fittings	Elbow (copper)	3"?	1	FT-06	VERIFY
Fittings	Elbow (jacketed)	—	3	FT-08	
Fittings	Flange pair	6"	4	FT-01 FT-05	
Fittings	Flexible connector	6"	1	FT-05	
Fittings	Press coupling	1"	1	FT-04	
Fittings	Press coupling	1-1/4"	1	FT-04	
Fittings	Process flange (sensor module)	—	1	FT-11	
Fittings	Rigid coupling	2-1/2"	3	FT-02	
Fittings	Tee	1-1/4" × 3/4"	1	FT-04	
Fittings	Tee (assumed)	1" × 3/4"	1	FT-04	VERIFY
Fittings	Tee (brass, crimp)	3/4"?	1	FT-09	
Fittings	Tee (grooved)	2-1/2"	1	FT-02	
Fittings	Tee (insulated)	4"	1	FT-03	
Fittings	Union (brass): copper → galvanized	3/4"?	2	FT-07	
Fittings	Welded branch (tee)	1"?	1	FT-07	
Hangers, supports & penetrations	Clevis / band (verify)	—	1	FT-06	VERIFY
Hangers, supports & penetrations	Insulation shield at support	8"	2	FT-03	
Hangers, supports & penetrations	Penetration — firestop sleeve	3/4"	3	FT-04	
Hangers, supports & penetrations	Penetration: sheet-metal sleeve	—	1	FT-10	
Hangers, supports & penetrations	Pipe clamp at strut	2-1/2"	1	FT-02	
Hangers, supports & penetrations	Pipe clip	—	6	FT-07	
Hangers, supports & penetrations	Pipe clip (typ.)	—	1	FT-07	
Hangers, supports & penetrations	Strut channel (stand)	—	1	FT-11	
Hangers, supports & penetrations	Trapeze strut post	1-5/8"	2	FT-03	
Hangers, supports & penetrations	Vertical member (verify)	—	1	FT-02	VERIFY
Hangers, supports & penetrations	Wall strut channel	1-5/8"	2	FT-02 FT-04	

Lengths are taken from the as-built model drafted from each photo, or from a confirmed pipe size used as a scale, with estimated dimensions. Rejected markups are left out. Items under 70% confidence are counted and flagged. Not for bidding. This shows how the takeoff would roll up.